

ORIGINAL ARTICLE

TVFNet: text and visual attention feature fusion network for multi-lesion segmentation of diabetic retinopathy

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Abstract

Diabetic retinopathy (DR) is a significant cause of blindness in the global population. Automatic and accurate lesion segmentation helps to assist ophthalmologists in DR diagnosis and grading. However, multiple lesion segmentation is full of challenges due to the inter-class similarities, intra-class differences and complex structures. Although previous research has made some progress, most works ignored the multi-source information of textual labels and visual features. This paper proposes a text and visual attention feature fusion network for multi-lesion segmentation of diabetic retinopathy. The proposed model incorporates textual labels of the number and scale of lesions with visual features by element-wise multiplication for multi-source information fusion. First, ResNeXt-50 is used to extract features from retinal images and classify lesion types and scales. Then, the feature enhancement module (FEM) adopts the triple attention mechanism composed of channel attention, spatial attention and pixel attention to focus on important lesion regions. Next, the text label attention module (TLAM) utilizes byte-pair encoding and element-wise multiplication to adaptively fuse textual label information with visual features as supervision for the decoder. Finally, the multi-scale feature fusion module (MSFM) adopts 1×1 convolution and sigmoid operation to generate lesion segmentation results. Extensive experimental results on four public datasets of IDRiD, DDR, FGADR and E-ophtha well demonstrate the superiority of our model, which achieves competitive performance compared to state-of-the-arts. It has better accuracy and robustness, which can overcome adverse interference from other fundus tissues and background noise for precise multiclass lesion segmentation.

Keywords Diabetic retinopathy · Multiclass lesion segmentation · Multi-source information fusion · Text label attention · Feature enhancement

1 Introduction

Diabetic retinopathy (DR) is one of the main complications of diabetes, which has become the leading cause of blindness in the global population [1, 2]. According to statistics from the International Diabetes Federation, approximately 425 million people suffer from diabetes worldwide. Moreover, the number of patients with diabetes will reach 592 million by 2035, of which 1/3 of them will suffer from DR [3]. Different pathological features appear during the development of DR, mainly including microaneurysms (MA), hemorrhage (HE), soft exudate



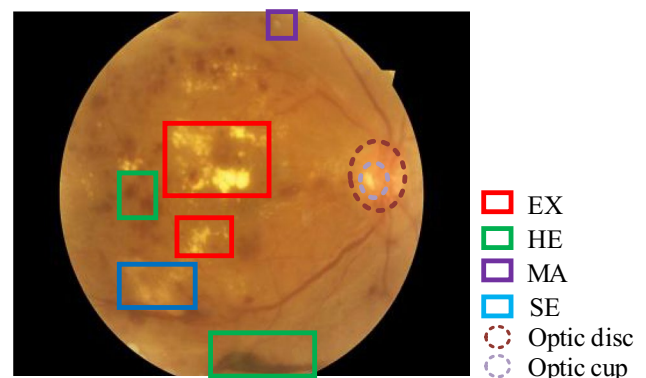
(SE) and hard exudate (EX), as shown in Fig. 1. MA often appear in the early stage of DR. Because of the expansion of microvessels and the increased vulnerability of vascular walls caused by diabetes, they appear as small round or oval small aneurysms with light color, which appear singly or in clusters. As the disease progresses, EX and SE with a few hemorrhagic dots usually appear in the middle stage of DR. EX is a bright yellow white plaque or spot formed by the overflow of lipoproteins from plasma to retinal tissue after vascular injury. The shape of soft exudate is irregular with blurred boundaries. It usually lights in color, and the surface presents a cotton like or feather like appearance. They are one of the indicators of DR progression, and severe exudate can lead to visual impairment. In the late stage of DR, new blood vessels begin to grow due to damage to retinal blood vessels, but these vessels are usually fragile and abnormal, which is prone to rupture and bleeding. The severity of DR is graded into five stages including normal, mild, moderate, severe and proliferative based on lesions, according to International Clinical Diabetic Retinopathy Disease Severity Scale [4, 5]. DR is a chronic progressive disease that affects vision and even causes blindness. Therefore, early detection and timely intervention can effectively curb the progress of diabetic retinopathy.

Due to the unique advantages of convenience and economy, two-dimensional color fundus photography (CFP) are often used as a main tools for the diagnosis and treatment of retinal diseases. Currently, ophthalmologists manually examine CFP for DR detection in real clinical practice. However, manual annotation of multiclass lesions by ophthalmologists is time-consuming and laborious, making it difficult to ensure the accuracy of diagnosis and widespread promotion and application. Therefore, it is necessary to design an automatic lesion segmentation method to reduce clinical workload and improve the accuracy and efficiency of ophthalmologists in diagnosing diabetic retinopathy.

Although significant achievement has been made in multiclass lesion segmentation, it is still a challenging issue due to the following four aspects. (1) The distribution of lesions is relatively sparse and complex. There are great differences in scale, shape, color, texture and brightness among various types of lesions, with no fixed pattern. (2) Some lesions (e.g., MA) are small in size and easily be confused as noise. (3) Intra-class similarity and inter-class variability exist between different lesions, which easily causes misclassification. (4) The appearances of fundus images from diverse datasets vary due to differences in lighting conditions and camera systems.

Recently, deep learning models are widely applied in medical image segmentation and disease analysis. The existing DR lesion segmentation methods are classified into encoder-decoder and non-encoder-decoder models. Some researchers cropped the original retinal images into patches and inputted them into U-Net [6] and its variant networks [7–9] for multi-lesion segmentation [10–13]. Although this method can effectively reduce the memory burden, it is prone to losing detailed information of small-scale lesions in downsampling. Other works [14–18] directly fed the whole images into non-encoder-decoder networks to fuse multi-level lesion features. It can preserve the detailed features of lesions, but ignores the inherent relationship between visual features and text contextual information. In fact, text label and visual features contain multimodal information, and fully integrating the two can help improve the performance of lesion segmentation. The previous studies focus more on the design

Fig. 1 Illustration of fundus image with DR multi-lesions



of network structure, which increases the complexity of the model. However, they rarely consider integrating text labels and visual features related to lesion scale and the number of lesion types for DR lesion segmentation.

Based on the above analysis, this paper proposes a text and visual attention feature fusion network for DR multiple lesion segmentation. The model adopts an encoder-decoder structure, where the input is original fundus images and the output is multi-lesion segmentation results. Unlike previous methods that only use visual features of fundus images, the proposed method introduces text-guided attention to fuse text labels with visual features of the lesion scale and the number of lesion types. Firstly, the original fundus images are fed to ResNeXt-50 to extract multi-level features. Secondly, the feature enhancement module (FEM) uses spatial attention, channel attention and pixel attention mechanisms to pay attention to important lesion areas. Thirdly, the text label attention module (TLAM) utilizes byte pair encoding and element-by-element dot product for adaptive fusion of text label information and visual features of fundus images as supervision for decoder. Fourthly, the multi-scale feature fusion module (MSFM) adopts bilinear up-sampling to fuse decoder features from different layers. Finally, the multi-scale feature fusion module uses 1×1 convolution and Sigmoid operations to generate lesion segmentation results. We evaluated the performance of TVFNet on four public datasets-IDRiD, DDR, FGADR and E-ophtha. Comprehensive comparison experiments and ablation studies show that the proposed model has good accuracy and robustness. It can overcome the adverse interference of similar tissues and background noise to achieve accurate segmentation of multiple lesions. To our best knowledge, this is the first time that combines text label information and visual features for multi-lesion segmentation of diabetic retinopathy.

The main contributions of this paper are as follows.

(1) To fully integrate the text label information and visual features of lesion scale and the number of lesion types, this paper proposes a text label attention module, which adopts byte pair encoding of text label and dot product of classification probability element by element for multimodal information fusion.

(2) To avoid losing small-scale lesion detail information in down-sampling, this paper proposes a feature enhancement module that uses a triple attention mechanism based on spatial attention, channel attention and pixel attention to capture important feature representations.

(3) We conduct extensive experiments on four public fundus image datasets. The ablation studies show that FEM, TLAM and MSFM contribute to accurate segmentation of multiple lesions. The comparison results demonstrate that our method has good accuracy and robustness, which achieves better performance compared to the state-of-the-art models.

The remainder of this paper is organized as follows. Section 2 briefly reviews current works on DR lesion segmentation. Section 3 provides the detailed introduction of the proposed network. Section 4 introduces implementation details, experimental data as well as results comparison and analysis on various datasets. Section 5 discusses the advantages and limitations of our model compared to the current works. Section 6 draws conclusions and suggests future research directions.

2 Related works

In recent years, multiple lesion segmentation of DR has raised public concern around the world, and many researchers have adopted deep learning model for lesion segmentation. According to the different features used, existing methods are classified into encoder-decoder networks and non-encoder-decoder networks.

2.1 Encoder-decoder based method

Due to the lightweight framework and ability to fully integrate deep and shallow features of U-Net, as well as the fixed structure and few shot of medical images, the encoder-decoder network performs superior in medical image segmentation. Therefore, some researchers adopted encoder-decoder models for DR lesion segmentation. Foo et al. [10] proposed a multi-task learning network for DR lesion segmentation and severity grading on the Kaggle

and IDRiD [19] datasets. The model uses VGG16 [20] as the encoder of U-Net, and the decoder uses transposed convolution instead of the upsampling operation, and transfers features to the decoder through skip connections before each maximum pooling. But the method did not explicitly segment the four types of lesions, providing only Dice scores for the segmentation of red and yellow lesions. Zhou et al. [11] proposed a semi-supervised collaborative learning method to simultaneously improve the performance of lesion segmentation and grading of DR. They adopted an Xception-module based U-Net to get various lesion masks, and utilized attention mechanism to generate the attention map based on the lesion mask for DR grading. However, this method uses dilated convolution to calculate the lesion features of each channel, and dilated convolution loses the detailed information of small lesions through rough feature down-sampling, making it difficult to preserve the fine edge details of the lesion area. Wang et al. [12] proposed a deep multi-task learning network that simultaneously improves the resolution of fundus images, DR lesion segmentation and grading. It uses a gradient-based multi-scale residual U-Net visualization algorithm to segment lesion areas. Although the method achieves better segmentation results on the four lesion segmentations on IDRiD and EyePACS, it only utilizes image features not fused with textual information on lesion type and scale for lesion segmentation. Guo et al. [13] proposed a cascaded attention RefineNet for multi-lesion segmentation in diabetic retinopathy. It takes the patch image and whole fundus image as dual inputs to the model, which integrates the fine local detail features and coarse global information for accurate lesion segmentation. However, the model misclassified noise as MA in fundus images with very low contrast, which may lead to false positive results. Pavani et al. [21] proposed RILBP-YNet for simultaneous multi-lesion segmentation. They added parallel encoders to encoder-decoder framework for introducing texture features in lesion segmentation.

Although the encoder-decoder network can fuse multi-level features using skip connections, its deconvolution operation cannot restore low-level edge details lost after downsampling, which leads to generate rough segmentation results.

2.2 Non-encoder-decoder based method

Considering the limitation of the encoder-decoder model, other researchers utilized non-encoder-decoder networks such as VGG or ResNet for multiple lesion segmentation in diabetic retinopathy. He et al. [14] proposed a progressive multi-scale consistent network (PMCNet) for DR multi-lesion segmentation, using dynamic attention blocks and progressive feature fusion modules to address the issues of conflicts between low-level and high-level features, as well as inconsistent lesion size and scale. Although this method performs well in segmenting the lesion areas of most images, it is difficult to accurately segment small MA and EX when dealing with low contrast and low-quality fundus images. Due to the spatial relationship between different lesions and vessels of fundus image, Huang et al. [15] proposed a relation Transformer network (RTNet) for multi-lesion segmentation. It adopts global Transformer module to obtain global features, and uses the local relationship Transformer module to model the spatial relationship between vessels and lesions using self-attention and cross-attention. The method can effectively capture implicit relationships between lesions and blood vessels, but it may result in boundary blurring when segmenting large-scale HE area. Wang et al. [16] proposed a contextual and local collaborative framework for multi-lesion segmentation of diabetic retinopathy. It utilizes attention mechanism to fuse local details and contextual information, and uses SE-ResNeXt-50 for lesion classification. However, it is difficult to detect small lesions and lacks large-scale clinical validation, which cannot guarantee the generalization of the model.

Zhou et al. [17] built a DR image dataset called FGADR with refined annotations, which includes 1842 pixel-level lesion labels and 1000 image-level grading labels. They compared the experimental performance using U-Net and non-U-Net networks on this dataset. The experimental results indicate that the segmentation performance is best when using Dense U-Net and U-Net++. Guo et al. [18] proposed L-Seg, which uses the pretrained VGG16 as the backbone network and combines multi-scale feature fusion module to segment four lesions simultaneously. Moreover, multi-channel cross entropy loss function is used to solve the class imbalance problem.

However, this method can only alleviate the issue of lesion misclassification to a certain extent, and it is difficult to accurately identify small-scale lesions in complex backgrounds. Xue et al. [22] proposed a dynamic membrane system with hybrid structures for MA and EX segmentation on Messidor [23], E-oPhtha [3] and IDRiD datasets. Guo et al. [24] proposed multi-scale network for DR lesion segmentation. They adopted fully convolutional neural networks (FCN) to extract multi-scale lesion features from retinal images with different resolutions. Then, an auxiliary CNN is applied to fuse the outputs of multiple FCNs. However, it does not fully consider the inter-class similarity of different lesions, so the overall generalization performance of the network needs to be improved.

Although the above methods can achieve good segmentation results on a specific lesion, most of them use fixed fusion weights for feature maps of different levels, ignoring the content differences between them. In addition, previous works paid more attention to the design of network structures, without fully considering the connection between text labels and visual features of lesions.

2.3 Multi-modal information fusion method

In clinical applications, doctors usually refer to different types and modalities of images for comprehensive analysis and cross validation for accurate diagnosis. Therefore, it is helpful for accurate diagnosis of the disease to adopt an effective multimodal information fusion strategy to fully explore and utilize the complementary information between different modal images. In recent years, the latest applications of multimodal fusion in the medical field mainly focus on the assisted diagnosis of brain and lung tumors, prognosis prediction, and diagnosis of fundus diseases.

Zhou et al. [25] proposed a Transformer based multimodal representation learning model for disease diagnosis, which learns the overall representation of multimodal data such as medical images, unstructured chief complaints, and structured clinical information. They used embedding layers to convert images, unstructured and structured text into visual and textual labels, and then utilized inter-pattern and intra-pattern bidirectional attention mechanisms to encode and learn the connections between different modalities. The method performs well on the lung disease diagnosis and COVID-19 recognition tasks. However, it requires much training data and cannot effectively handle unregistered words when using word embeddings for text encoding. Shi et al. [26] proposed a visual-language foundation model for detection of various fundus diseases. They adopted a pre-training strategy that integrates self-supervised reconstruction, multimodal image contrastive learning, and image text contrastive learning to explore prior information in 2.77 million multimodal ophthalmic images and 110000 text data, and learn shared feature representations of different modalities.

Li et al. [27] fused segment anything model (SAM) with explicit text information for DR lesion segmentation. This method utilize text description prompts for different types of lesions instead of labels to focus on lesion location, transforming implicit medical concepts into explicit prior knowledge to provide more interpretable clues and explore low-level lesion features. Then prior knowledge is injected to accelerate knowledge sharing between multimodal features, to improve the efficiency of model computation iteration. However, due to the lack of corresponding lesion text description labels in most DR datasets, manual labeling is time-consuming and labor-intensive. Moreover, visual language models rely on natural image training and face semantic alignment issues. Aleem et al. [28] used SAM to segment the original image into multiple areas, and then retrieved regions of interest (ROI) from many masks generated by SAM through contrastive language-image pre-training (CLIP). Finally, the retrieved ROI was used as the prompt to guide SAM segmentation of the target organ. However, it ignores that the semantic representations extracted from vision language models (VLMs) are mostly abstract and non-pixel level, which inevitably introduces noise. In addition, it is difficult to accurately align with image features and rough text descriptions, which may reduce segmentation accuracy. Li et al. [29] proposed a dual U-shaped LViT model for lung image segmentation. The convolutional neural network branch serves as information input and the segmentation head of prediction output, while the ViT branch uses embedding layers instead of text encoders to calculate text feature vectors and merge image and text information. The pixel level attention

module preserves local features of the image and merges semantic features in the text by fusing channel attention and spatial attention mechanisms.

To overcome the limitation of previous methods, this paper presents a text and visual attention feature fusion network for DR multiple lesion segmentation. The differences between our method and current works are shown in the following two aspects. (1) To prevent losing detailed information during the downsampling stage, our model proposes a feature enhancement module that uses a triple attention mechanism to retain more lesion details, which makes network pay more attention to tiny and inconspicuous lesion area. (2) To fully combine fundus visual features and text context information, the proposed network proposes a text label attention module. It uses byte-pair encoding of text labels and dot product of classification probability element by element for multimodal information fusion.

3 Methodology

Figure 2 shows the overall framework of the proposed model. It adopts an encoder-decoder architecture, consisting of encoder, feature enhancement module, text label attention module, multi-scale feature fusion module and decoder. The encoder uses ResNeXt-50 [30] as backbone to extract the whole fundus image features and obtain the classification probabilities for the number of lesion types and the scale of lesions. The feature enhancement module (FEM) utilizes four parallel dilation convolutions and a triple attention mechanism to capture lesion features for enhancing the explicit relationships between different types of lesion spatial positions, channels and pixels. The text label attention module (TLAM) multiplies classification probability of the number of lesion types and lesion scale with the text label information in an element-by-element dot product as supervision for feature

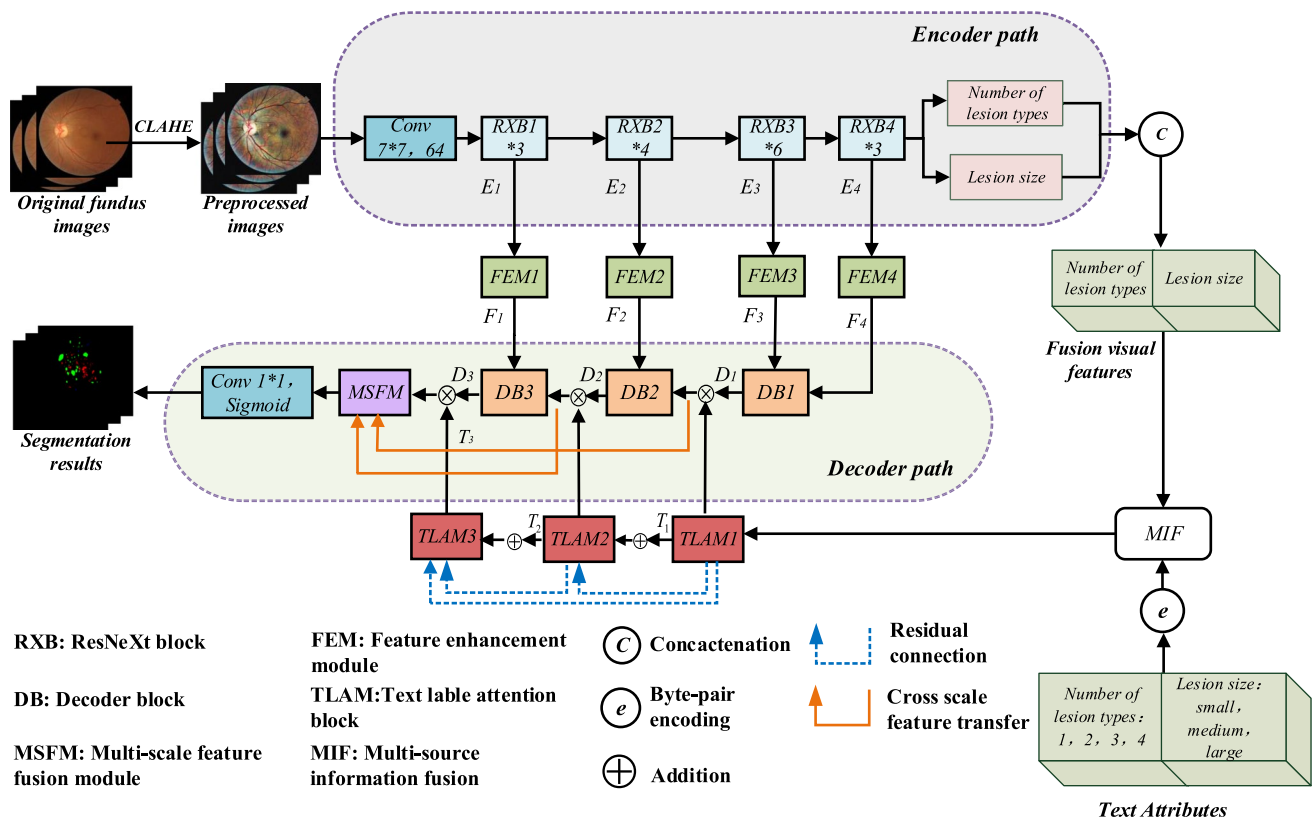


Fig. 2 Overall architecture of proposed TVFNet framework

upsampling. The multi-scale feature fusion module (MSFM) utilizes a series of convolutions, batch normalization and ReLU operations to fuse multi-scale lesion features through residual connections.

3.1 Overview of TVFNet

3.1.1 Encoder

The encoder uses ResNeXt-50 [30] pretrained on ImageNet [31] as the backbone network for feature extraction. ResNeXt-50 includes four residual blocks, and the resolution of output after each residual block is reduced to 1/4 of the original image. Figure 3 displays the structure of the ResNeXt block, it follows the split-transform-aggregate strategy and performs group convolution operations through 32 paths. Each path of the ResNeXt block contains three convolution layers, two batch normalization layers and three ReLU functions. ResNeXt utilizes parallel stacked grouped convolutional residual blocks with the same topology structure to extract multi-level lesion features, which effectively improves the accuracy of model without significantly increasing the computational parameters.

ResNeXt-50 outputs the classification prediction probability of the lesion type number and scale by training two classifier. In the two classification tasks, the number of lesion type's labels are classified into four categories of 1, 2, 3 and 4, while the scale classification labels are divided into three categories of small, medium and large.

It is not clearly specify the scale of DR lesion in previous studies. The current work [32] classifies colon polyps into four categories based on their diameter. It is specified that tiny polyps with diameter less than 0.5cm, small polyps with diameter between 0.5cm and 1 cm, moderate polyps with diameter range of 1~2 cm, and large polyps with diameter greater than 2cm. Motivated by the clinical experience and the analysis of different fundus image datasets, the lesions are classified into three different scales of small, medium and large based on the number of pixels occupied by each lesion. The number of pixels less than 200 was set as small lesions, the number of pixels between 200 and 1000 as medium lesions, and larger than 1000 as large lesions.

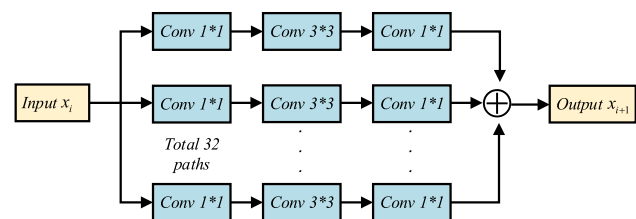
3.2 Feature enhancement module

3.2.1 Motivation

The encoder may lose some detailed information during the downsampling process, which is crucial for segmenting small-scale lesions. In addition, directly using skip connections to transfer the encoding features to the decoder can only fuse features of the same spatial resolution on a single scale, which limits the ability of the network to extract multi-level features. Therefore, it is necessary to design a feature enhancement module (FEM) to capture effective representation of multiple attention features.

Considering the above issues, we propose FEM based on multi-group dilation convolution and triple attention mechanism. First, we use four groups of dilation convolutions [33] with different dilation rates to enhance global semantic features. Moreover, we adopt channel attention, spatial attention and pixel attention mechanisms to capture multiple attention features, which makes the network pay attention to the lesion area of interest. The spatial attention module highlights the spatial position of lesions in the feature tensor by sharing weights. The channel attention module selects specific feature channel patterns by assigning different weights to multiple

Fig. 3 The structure of ResNeXt Block



feature channels, while the pixel attention module adjusts the attention weights of different feature points to focus on small-scale lesions. Therefore, designing this module is beneficial for dynamic feature enhancement and maintaining feature consistency.

3.2.2 Structure details

The overall structure of FEM is shown in Fig. 4. It takes the output features of ResNeXt block as input and uses dilation convolution and triple attention mechanism to enhance multi-level lesion feature. Firstly, FEM uses four parallel dilation convolutions to enlarge the receptive field of fundus image. The dilation rates are set to 1, 2, 4, 8, respectively. The batch normalization and ReLU function are followed after each dilation convolution. The series of 1×1 convolution, batch normalization and ReLU operations is referred to as the CBRM. Secondly, the output features of the CBRM are fed into the channel attention module to capture the explicit relationship between feature channels. Thirdly, the channel relationship features output by the four-channel attention modules are concatenated, and then perform 3×3 convolution and batch normalization operations. Fourthly, the outputs of ResNeXt-50 module are input to the pixel attention module for feature selection. The operation makes the network focus on significant lesion features. Finally, the channel attention fusion features and pixel attention features are added, and the spatial attention module is used to suppress irrelevant areas and strengthen the lesions.

The detailed structure of triple attention mechanism module is shown in Fig. 5. It includes channel attention module, spatial attention module and pixel attention module.

(1) Channel attention module: The output features of the CBRM are defined as $M_i \in R^{W \times H \times C}$, which W , H and C represent the width, height and number of channels of the feature map. First, $M_i \in R^{W \times H \times C}$ is reshaped to $M_i^{c1} \in R^{W \times H \times C}$ and $M_i^{c2} \in R^{W \times H \times C}$, respectively. Then, matrix multiplication and Softmax operations are used to get the channel attention map $t^c \in R^{C \times C}$. Finally, the final channel attention features is calculated by $C_i = M_i + \text{Reshape}(t^c M_i^{c3})$.

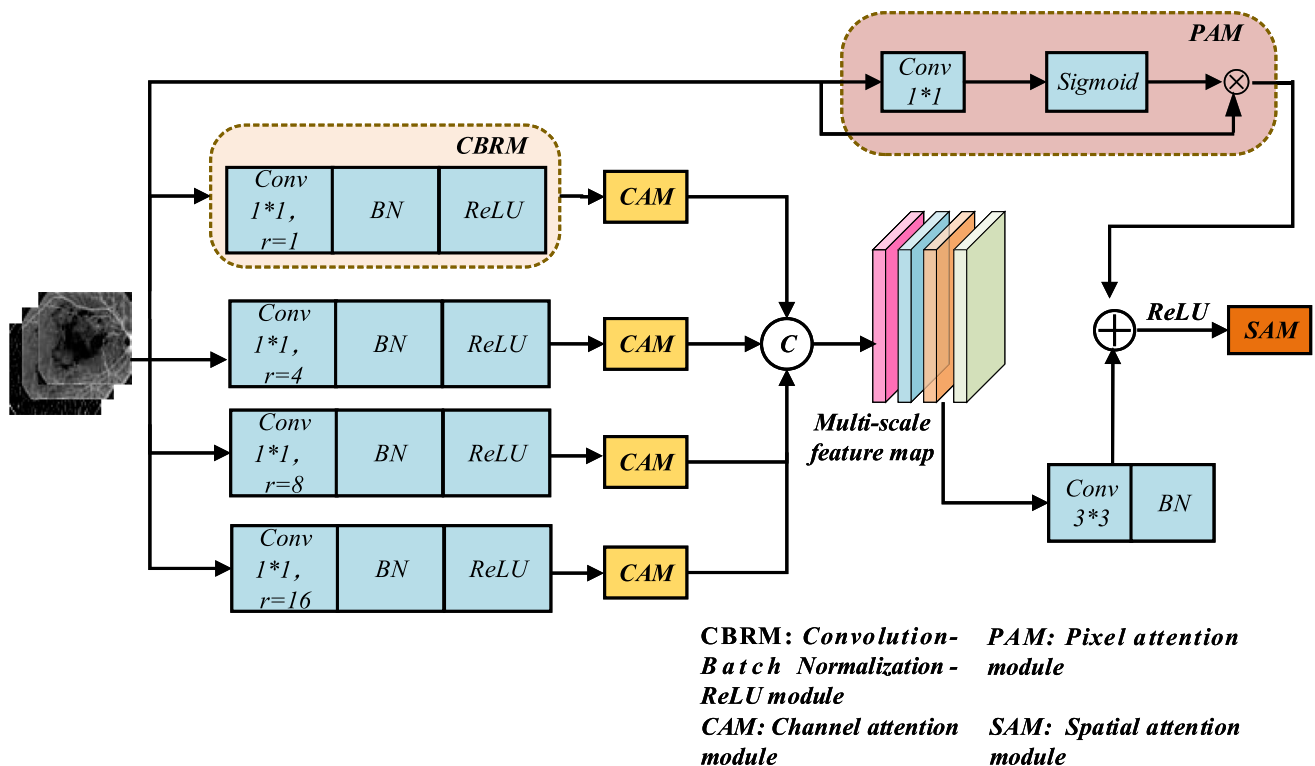
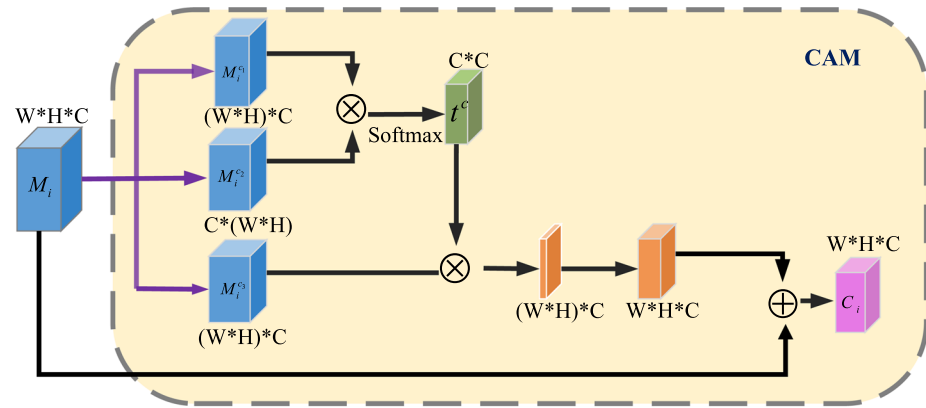
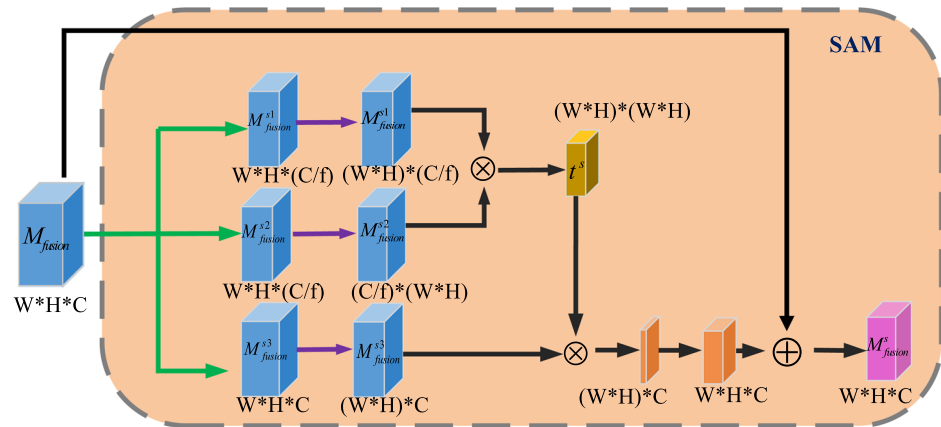


Fig. 4 The structure of feature enhancement block

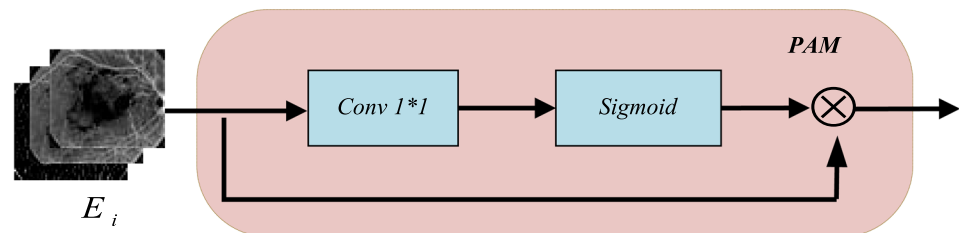
Fig. 5 Detailed structure of triple attention mechanism module. The purple arrow represents the reshape operation, the green arrow represents the convolution operation, and the black arrow represents the information flow



(a) The structure of channel attention module



(b) The structure of spatial attention module



(c) The structure of pixel attention module

(2) Spatial attention module: The three convolution operations are applied on the fused feature map of channel attention and pixel attention to get three branches. First, the channel dimension of feature map M_{fusion} on the first two branches is reduced to C/f to reduce computational complexity. Next, multiplication operation is performed on the transformed feature maps $M_{fusion}^{s1} \in R^{(W \times H) \times (C/f)}$ and $M_{fusion}^{s2} \in R^{(W \times H) \times (C/f)}$ of the first two branches. Then, Softmax function is adopted to obtain the spatial attention map $t^s \in R^{C \times C}$. Finally, the lesion spatial attention feature map is calculated by $M_{fusion}^s = M_{fusion} + Reshape(t^s M_{fusion}^{s3}).f$ as 4, 8, 16, and 32 to adapt to the feature maps with different resolutions of $28 \times 28 \times 2048$, $56 \times 56 \times 1024$, $128 \times 128 \times 512$ and $256 \times 256 \times 256$.

(3) Pixel attention module: We use 1×1 convolution and Sigmoid activation on the feature map of ResNeXt block to get pixel attention weights, and multiply them with original feature map E_i to get the pixel attention features. The pixel attention features are calculated by the following equation:

$$W_i = \sigma(\text{Conv}_{1 \times 1}(E_i)), \quad (1)$$

$$M_i^p = W_i \otimes E_i, i = 1, 2, 3, 4, \quad (2)$$

where σ represents the Sigmoid function. W_i represents attention weights with the same resolution as the input feature map, which is used to represent the importance of each pixel. M_i^p represents pixel attention feature map.

3.3 Text label attention module

3.3.1 Motivation

Most existing methods only utilize fundus image features and fail to effectively fuse textual contextual information with visual features for lesion segmentation of DR. Fundus images and text labels come from different information sources and contain features of different modalities. Therefore, it is helpful to locate lesion areas by integrating text label information with visual features of fundus images.

Considering the above issues, we propose a text label attention module (TLAM) to fully integrate features from different information domains. The labels of the number of lesion types and scale in the two classification tasks are encoded by byte-pair-encoding and multiplied with the classification probabilities pointwise for adaptive feature fusion.

3.3.2 Structure details

The TLAM takes the element-by-element dot multiplication of the number of lesion types and scale classification probability $\{\delta_{one}, \delta_{two}, \delta_{three}, \delta_{four}, \delta_{small}, \delta_{medium}, \delta_{large}\}$ and text encoding label as input. Let $A = \{one, two, three, four, small, medium, large\}$ represent the byte-pair encoding attribute, and $A_{encoder} = \{e_k^j, k = 1, \dots, 7\}$ represent the vector set of attributes k with length j , then the multi-source information fusion operation of text coding label and Softmax classification probability is expressed as:

$$T_i = \delta_k \odot e_k^j, \forall j. \quad (3)$$

where T_i represents the output features of TLAM. We perform 1×1 convolution, ReLU and 1×1 convolution operations on the output of the low-level TLAM, and transfer to the high-level TLAM through skip connections to fuse multi-level text information and visual features, as shown in Fig. 6.

3.4 Decoder

Figure 6 shows the specific structure of decoder. Each decoder block uses the output features of the TLAM as supervision for upsampling, and adopts a series of convolutions, batch normalization and ReLU functions to transfer the upsampling features to subsequent blocks to get multi-level lesion features.

First, the first decoder upsamples the output features F_4 using bilinear interpolation to increase the resolution to twice the original, and performs concatenate fusion with the output F_3 of the third FEM, followed by feeding the fusion feature to the CBRM. Next, multiple residual connections are used to perform ReLU linear activation, 3×3 convolution and batch normalization on the output features of CBRM, and fed into the convolutional block attention module (CBAM) [34] to obtain D_i^{cbam} . Then, the outputs of the TLAM are sequentially subjected to 1×1 convolution and ReLU operations, and multiplied with the output features of the decoder block to obtain multi-level lesion features in pixel-wise.

Therefore, the output of the $i - th$ decoder module is represented as:

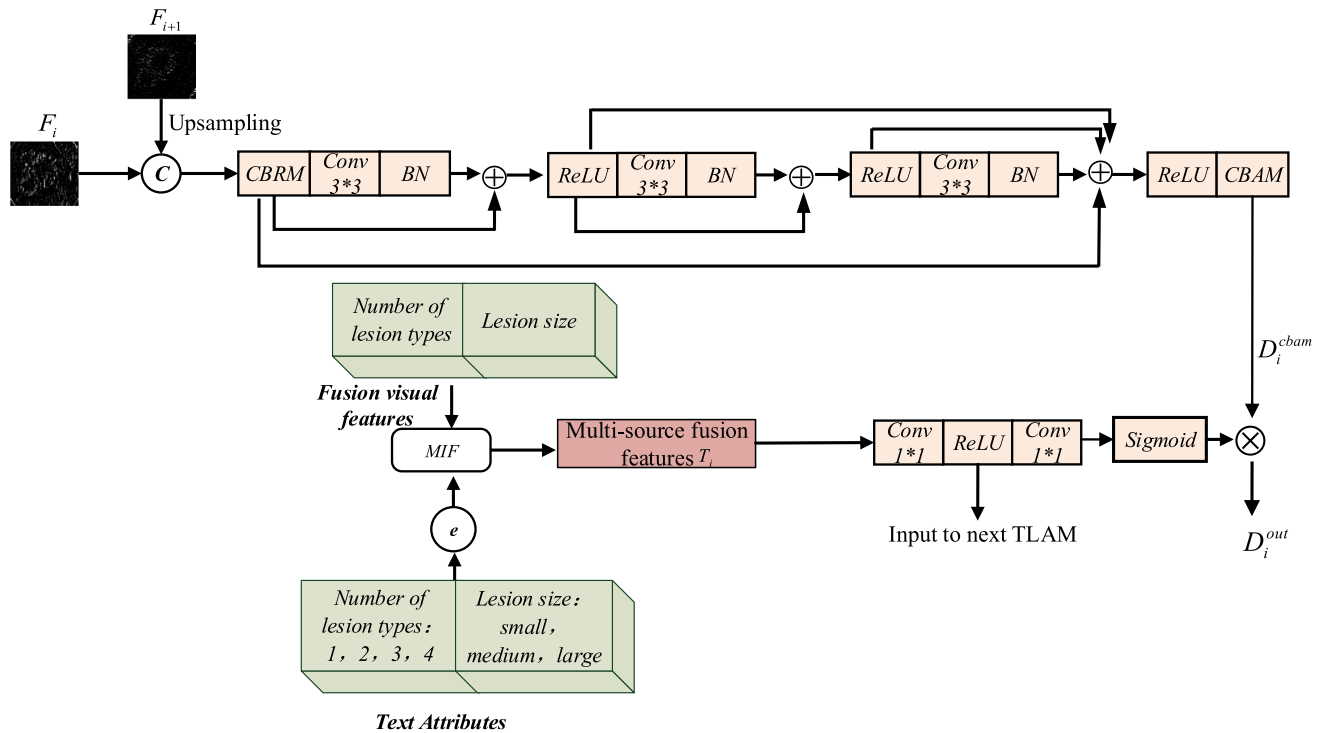


Fig. 6 The structure of decoder

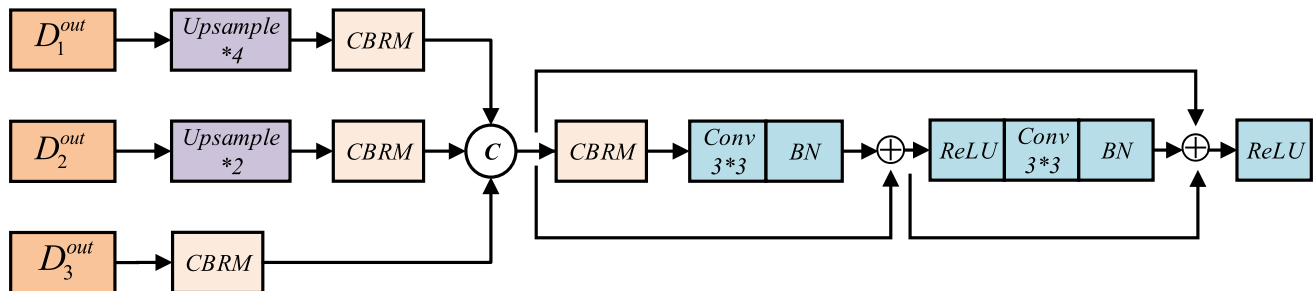


Fig. 7 The structure of multi-scale feature fusion block

$$D_i^{out} = D_i^{cbam} \odot \sigma(\text{Conv}_{1 \times 1} - \text{ReLU} - \text{Conv}_{1 \times 1}(T_i)). \quad (4)$$

where δ denotes Sigmoid function, $\odot$ denotes dot product in element-wise.

3.5 Multi-scale feature fusion module

The structure of the multi-scale feature fusion module (MSFM) is shown in Fig. 7, which fuses the output features of different decoders to get lesion features at multi-levels. Firstly, MSFM upsamples the output features of the first and second decoder blocks, and then performs 1×1 convolution, batch normalization and ReLU operation. Afterwards, the fused features are concatenated with the output features of the third decoder block. Then, a series of convolutions, batch normalization and ReLU activation is applied to the fused features, which uses residual connections to speed up information flow. The final lesion segmentation results are represented as:

$$P = \delta(\text{Conv}_{1 \times 1}(M_f)), \quad (5)$$

where M_f represents the output of MSFM, and δ represents the Sigmoid activation function.

3.6 Loss function

The proposed network adopts ResNeXt-50 for scale and lesion type classification, which results in two types of classification losses. Let $N = \{n_i^k, i = 1, 2, \dots, m, n_i^k \in (1, 2, 3, 4)\}$ represent the label of lesion type number, $C = \{c_i^k, i = 1, 2, \dots, m, c_i^k \in (0, 1, 2)\}$ represent the label of lesion scale, $P_N = \{p_{n_i}^k, i = 1, 2, \dots, m, p_{n_i}^k \in (1, 2, 3, 4)\}$ the classification prediction of lesion type, and $P_C = \{p_{c_i}^k, i = 1, 2, \dots, m, p_{c_i}^k \in (1, 2, 3, 4)\}$ denote the prediction of lesion scale. Therefore, the classification loss is represented as follows:

$$\begin{aligned} L_{cls} &= L_{CE1}(N, P_N) + L_{CE2}(C, P_C), \\ &= -\frac{1}{4} \sum_{i=1}^m \sum_{k=1}^4 n_i^k \log n_i^k - \frac{1}{3} \sum_{i=1}^m \sum_{k=1}^3 c_i^k \log c_i^k, \end{aligned} \quad (6)$$

where L_{CE1} and L_{CE2} both represent the cross-entropy loss.

Furthermore, the segmentation loss is calculated the difference between the labels and prediction of each pixel in the fundus image. However, due to the severe imbalance between healthy and lesion pixels, segmentation results will inevitably be biased towards normal pixels if not consider data imbalance issues. Therefore, we adopt the weighted sum of Dice loss and class-balanced binary cross-entropy loss as segmentation loss to address the potential overfitting issue. Let $G = \{g_i, i = 1, \dots, m, g_i \in (0, 1)\}$ represent the lesion labels corresponding to the fundus image, and $P = \{p_i, i = 1, \dots, m, p_i \in (0, 1)\}$ represent the segmentation predictions of algorithm. The segmentation loss is expressed as:

$$\begin{aligned} L_{seg} &= \mu_1 L_{BCE}(G, P) + \mu_2 L_{Dice}(G, P) \\ &= \mu_1 \cdot [-\eta g_i \log p_i - (1 - \eta)(1 - g_i) \log(1 - p_i)] \\ &\quad + \mu_2 \left(1 - \frac{2 \langle g_i, p_i \rangle}{\|g_i\|_1 + \|p_i\|_1}\right), \end{aligned} \quad (7)$$

where L_{BCE} and L_{Dice} represent the Dice loss and balanced cross-entropy loss function, respectively. μ_1 and μ_2 denote the balancing parameters used to balance the binary Dice loss and cross-entropy loss function. η denotes the weighting factor used to balance the difference between lesion and normal pixels. $\langle g_i, p_i \rangle$ denotes the element-wise multiplication between the manual label and predictions of each channel. $\|\cdot\|_1$ represents the L1 norm, which is the sum of absolute values of all matrix elements.

In summary, the goal of our network is to achieve rapid convergence by minimizing the weighted sum. The overall loss function includes classification loss and segmentation loss, and different balance factors are used to weight these two losses, which is represented as:

$$L = \gamma_1 L_{cls} + \gamma_2 L_{seg}, \quad (8)$$

where γ_1 and γ_2 represent the balance factors used to balance the classification loss and segmentation loss, respectively.

4 Experiments and results

4.1 Datasets

We conducted experiments on four public CFP datasets of IDRiD, DDR, FGADR-Seg-set and E-ophtha to validate the performance of the proposed model. The general information of each dataset is shown in Table 1.

IDRiD [19]: It contains 81 fundus images of Indian individuals with a resolution of 4288×2848 pixels, each image has expert manual annotations of four lesions. The ratio of training and test images in IDRiD dataset was set to 7:3 in the experiment, 54 images were used for training and 27 images were used for testing. The dataset is openly available at <https://idrid.grand-challenge.org/>.

DDR [35]: It includes 757 retinal images with pixel-level labels for four types of lesions from Chinese individuals. The resolution of retinal images varies from 1380×1382 to 2736×1824 . The ratio of training, validation and test images in DDR dataset was set to 5:2:3 in the experiment, 383 images were used for training, 149 images used for validation, 225 images used for testing. The dataset is openly available at <https://github.com/nkicsl/DDR-dataset>.

FGADR [17]: It contains 2828 fundus images of individuals from the United Arab Emirates, including two sub-datasets for segmentation and grading. The Seg-set includes 1842 retinal images, each of which has pixel-level labels for multiple types of lesions. The ratio of training, validation and test images in FGADR dataset was set to 7: 2:1 in the experiment, 1289 training images were used for training, 368 images used for validation, 185 images used for testing. The Grade-set consists of 1000 fundus images with grading labels provided by six ophthalmologists. The dataset is openly available at <https://csyizhou.github.io/FGADR/>.

E-ophtha [3]: It includes 463 fundus images of French individuals, which consists of two sub-sets: E-ophtha_MA and E-ophtha_EX. E-ophtha_MA contains 381 fundus images with corresponding manual labels, of which 233 are normal images and 148 are MA images. The ratio of training and test images in E-ophtha_MA was set to 7:3 in the experiment, 287 images were used for training and 94 images were used for testing. E-ophtha_EX includes 82 retinal images with pixel-level manual annotations, of which 35 are normal images and 47 are EX images. The ratio of training and test images in E-ophtha_EX was set to 7:3 in the experiment, 52 images were used for training and 30 images were used for testing. The dataset is openly available at <https://www.adcis.net/en/third-party/e-ophtha/>.

4.2 Implementation details

All models were implemented using pytorch 1.7.1 framework and performed on NVIDIA GeForce RTX 2080 GPU with 11GB of memory. ResNeXt [30] pretrained on ImageNet [31] was used as backbone of encoder. We adopted ADAM [36] for rapid convergence of the network, and applied Poly learning strategy to iterate the learning rate in the training stage. The iteration way of learning rate is to multiply the initial learning rate by $(1 - \frac{epoch}{max_epoch})^{power}$, with the initial learning rates set to 10^{-4} , and β_1 and β_2 set to 0.9 and 0.99, respectively, and ε set to $1e-7$. The momentum *power*, batch size and maximum number of iterations *max_epoch* were set to

Table 1 Basic information of experimental datasets

Dataset	Total number	HE	MA	SE	EX	Healthy	Country
IDRiD	81	80	81	40	81	0	India
DDR	757	601	570	239	486	-	China
FGADR-Seg-set	1842	1456	1424	627	1279	-	United Arab
E-ophtha_EX	82	-	-	-	47	35	France
E-ophtha_MA	381	0	148	-	-	233	France

0.9, 6 and 200, respectively. The balance factors μ_1 , μ_2 and η in segmentation loss were set to 0.7, 0.3, and 0.35, respectively. The weighting parameters γ_1 and γ_2 in overall loss were set to 0.4 and 0.6, respectively.

Moreover, necessary data preprocessing operations were performed on the original images before feeding them into the proposed network for data augmentation and lesion enhancement. First, due to the different resolutions of the fundus images in various datasets and GPU memory limitations, the fundus images were uniformly resized to 1024×1024 pixels. Second, contrast limited adaptive histogram equalization was used to enhance the contrast between the background and lesion areas. Finally, data augmentation operations were performed due to limited samples. (1) Scaling the original images ranging from 0.6 to 1.2 in step of 0.1. (2) Rotating the original images from 0 to 360 degrees in step of 40 degrees. (3) Translating the whole images horizontally and vertically from -60 pixels to 120 pixels in steps of 30 pixels.

4.3 Evaluation metrics

Accuracy (*Acc*), *Dice*, AUC_PR and AUC_ROC are used to evaluate the lesion segmentation performance of different networks, which are calculated as follows:

$$Acc = \frac{TP + TN}{TN + TP + FP + FN}, \quad (9)$$

$$Dice = \frac{2TP}{2TP + FN + FP}. \quad (10)$$

Specifically, *Acc* represents the proportion of correctly predicted pixels to the total pixels. *Dice* represents the similarity between the model segmentation result and the ground truth. AUC_PR and AUC_ROC represent the area below the precision-recall and receiver operating characteristic curve. The closer the above curve is to the upper right corner of the coordinate system, the better the classification performance of the algorithm.

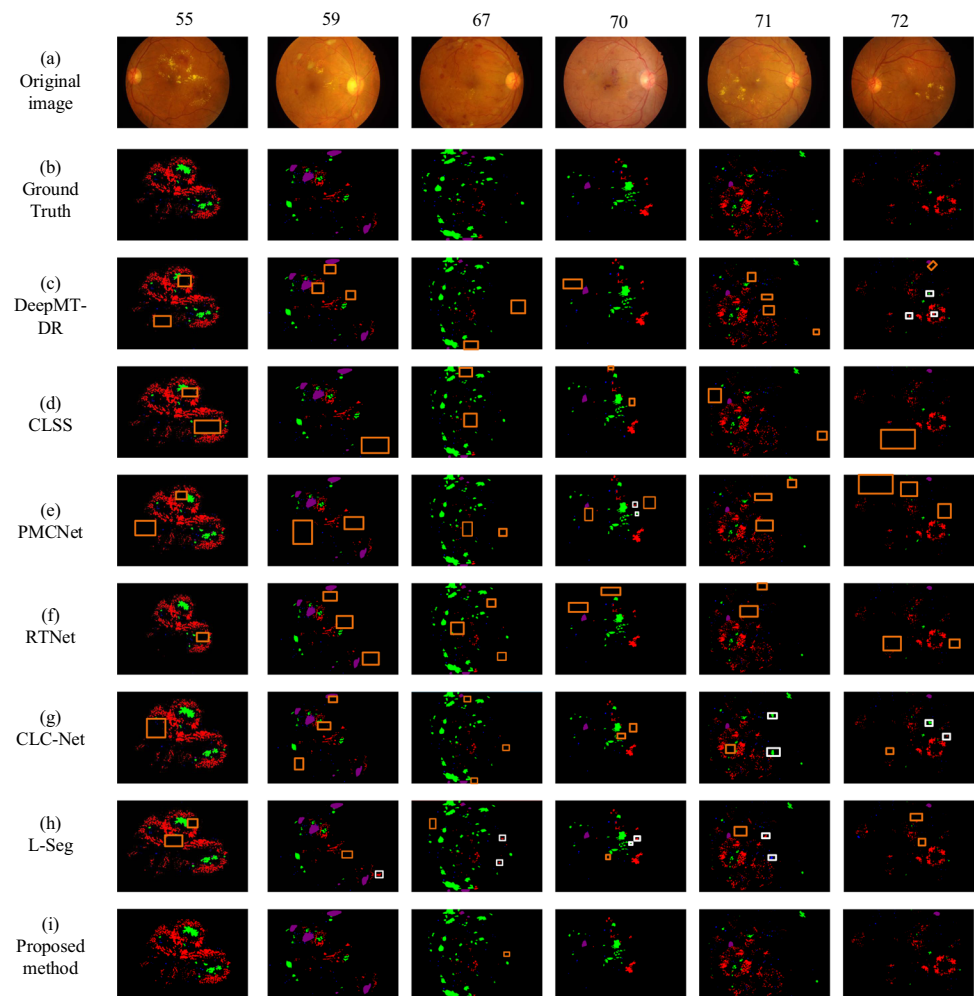
4.4 Comparisons with state-of-the-art methods

4.4.1 Performance on IDRiD

We compared the qualitative segmentation results of our approach with those of other existing networks [11, 14–16, 24] and Top three models on the IDRiD challenge to verify the effectiveness of the model. Figure 8 shows the visual segmentation results of eight different models on four types of lesions. Four different colors are used to denote different lesions, where red, green, blue and purple represent EX, HE, MA and SE, respectively. The numbers at the top of retinal images in Fig. 8 represent the specific sequence numbers of fundus images in IDRiD dataset. As shown in Fig. 8, the sizes and intensities of lesions in IDRiD are inconsistent, and the segmentation accuracy is affected by multiple spots. CLSS [11] used dilated convolution to extract lesion features and combined U-Net with Xception to generate segmentation masks of multiple lesions. However, dilated convolution introduces rough feature down-sampling, which leads to loss details of some small HE and MA. Although PMC-Net [14] can accurately segment large EX and HE, it ignored small MA and misclassified dark spots as hemorrhage. As different types of lesions have certain spatial correlations, RTNet [15] adopted global Transformer and local Transformer to segment EX and HE well. However, it may lose the contour details of small-scale MA and SE, resulting in blurry and coarse lesion segmentation results. CLC-Net [16] can accurately detect EX and SE, but it tends to misclassify MA as HE. L-Seg [24] is prone to background interference from bright noise and unable to distinguish hard exudates from drusens, which makes segmenting tiny EX in high-contrast fundus images difficult.

Compared with other methods, our method achieves better segmentation results. It can effectively overcome the adverse interference of background noise such as dark spots and artifacts, and preserve the edge details of

Fig. 8 Qualitative comparison results of various lesion segmentation methods on IDRiD. The orange and white bounding boxes represent false negatives and false positives, respectively



small-scale lesions. The main reason is that TVFNet first uses ResNeXt-50 to obtain the classification probability of the number and size of lesions. Then, FEM captures various attention features using triple attention mechanism to enhance the explicit relationship between spatial positions, feature channels and pixels of different types of lesions, which can focus on the lesion areas of interest. Moreover, TLAM performs element-wise multiplication of the classification probability of the number and size of lesions with the byte-pair encoding text label to fully integrate visual features and textual context features. In summary, our method adopts an encoder-decoder architecture, and integrates FEM and TLAM to focus on the important lesion areas and adaptively fuses multiple source features, which helps achieve precise segmentation of multi-scale lesions of different types.

Table 2 displays the quantitative comparison results of various models on the IDRiD dataset. As can be seen from Table 2, the segmentation performance of our network is better than most existing methods, achieving accuracies and Dice values of 0.9513, 0.9423, 0.9386, 0.9276 and 0.9525, 0.9575, 0.9239, 0.9171 on segmentation of EX, SE, HE, and MA, respectively. Compared with CLC-Net [16], AUC_PR and AUC_ROC of our method increased by 7.45%, 16.51%, 14.97%, 15.21% and 4.42%, 4.06%, 4.96%, 1.82%, respectively. The *Dice* and AUC_PR of our method increased by 24.13%, 15.63%, 32.99%, 47.91% and 13.57%, 3.67%, 18.59%, 16.62% when compared with KLD [37].

Figure 9 shows the ROC curves of four types of lesion segmentation on IDRiD dataset for five different models. The AUC values of EX, SE, HE and MA segmentation reach 0.9422, 0.9516, 0.9407 and 0.9124, respectively, which are higher than that of other four models. As shown in Fig. 9 (a) and (b), the AUC value of our method is the largest, followed by RTNet [15], and the AUC value of DeepMT-DR [12] is the smallest. It is observed from

Table 2 Quantitative comparison results of different models on IDRiD. (The best results are shown in bold)

Methods	EX			SE			HE			MA		
	Acc	Dice	AUC_PR	AUC_ROC	Acc	Dice	AUC_PR	AUC_ROC	Acc	Dice	AUC_PR	AUC_ROC
MTUnet [10]	0.8416	0.8427	-	0.9327	0.8367	0.8423	-	0.9107	0.8321	0.7515	-	0.8874
CLSS [11]	0.8349	0.8568	0.8872	0.9833	0.7054	0.8143	0.7407	0.9536	0.6983	0.7312	0.6936	0.9328
DeepMT-DR [12]	0.8469	0.8423	0.8861	0.7835	0.7415	0.8019	0.7051	0.8689	0.8429	0.8983	0.9012	0.7907
CARNet [13]	0.9283	0.9521	0.8987	0.9768	0.9512	0.9542	0.8739	0.9529	0.9257	0.9183	0.7929	0.9117
PMCNet [14]	0.9127	0.9513	0.8724	0.8234	0.8852	0.9215	0.6705	0.8855	0.9124	0.8973	0.7111	0.8645
RTNet [15]	0.9352	0.9417	0.9024	0.9103	0.8983	0.9017	0.6880	0.8943	0.9141	0.8479	0.4897	0.8671
CLC-Net [16]	0.8745	0.9152	0.8272	0.8024	0.8412	0.9153	0.6611	0.7108	0.9032	0.7983	0.6722	0.8117
L-Seg [18]	0.9452	0.9478	0.7945	0.9429	0.9238	0.9089	0.7113	0.9319	0.8973	0.7986	0.6371	0.8369
RILBP-YNet [21]	0.9316	0.9047	0.8644	-	0.8967	0.9128	0.7597	-	0.7985	0.9012	0.6411	-
DMS [22]	0.9460	0.7790	0.8970	0.7640	-	-	-	-	-	-	-	0.6720
VRT	0.9319	0.9327	0.7127	-	0.9217	0.9398	0.6995	-	0.8896	0.9206	0.6804	-
PATech	0.9225	0.9302	0.8850	-	0.9178	0.9235	0.7051	-	0.9019	0.9012	0.6490	-
iFLYTEK-MIG	0.9137	0.9069	0.8741	0.9339	0.9099	0.9217	0.6588	0.9213	0.8893	0.8798	0.5588	0.9225
LKD [37]	-	0.7112	0.7660	-	-	0.8012	0.7985	-	-	0.5940	0.6360	-
HACDR-Net [38]	-	0.7850	0.8690	-	-	0.7196	0.7673	-	-	0.6427	0.6850	-
MSGDA-Net [39]	-	0.5562	0.6120	-	-	0.5566	0.6053	-	-	0.5561	0.6051	-
M2MRF [40]	-	0.7997	0.8229	-	-	0.6792	0.6900	-	-	0.6642	0.6869	-
DMSF [24]	-	0.8952	0.8961	-	-	0.9230	0.9028	-	0.5698	0.6985	0.6884	0.9012
Proposed	0.9513	0.9525	0.9017	0.9566	0.9423	0.9575	0.8262	0.9516	0.9386	0.9239	0.8219	0.9124

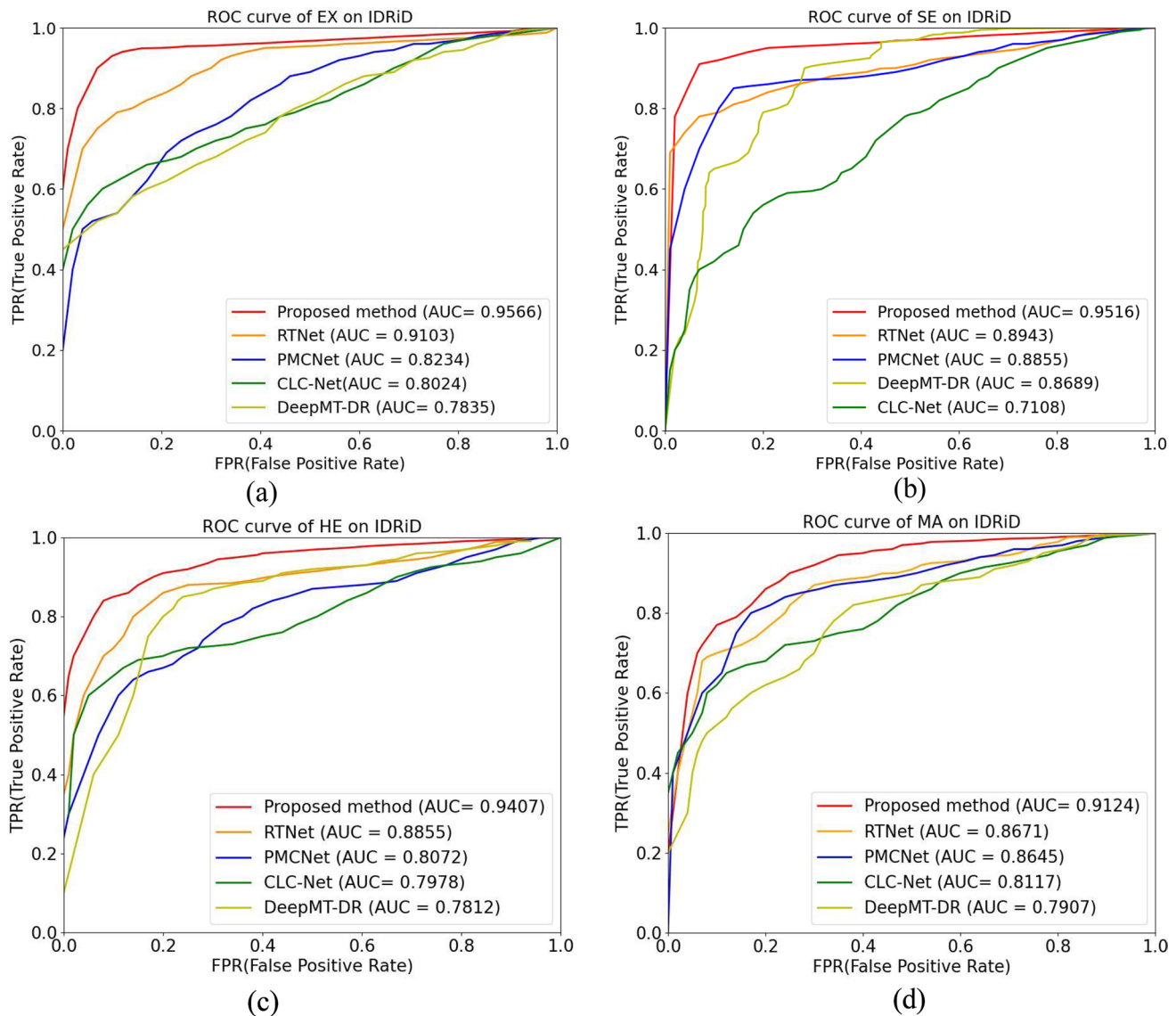


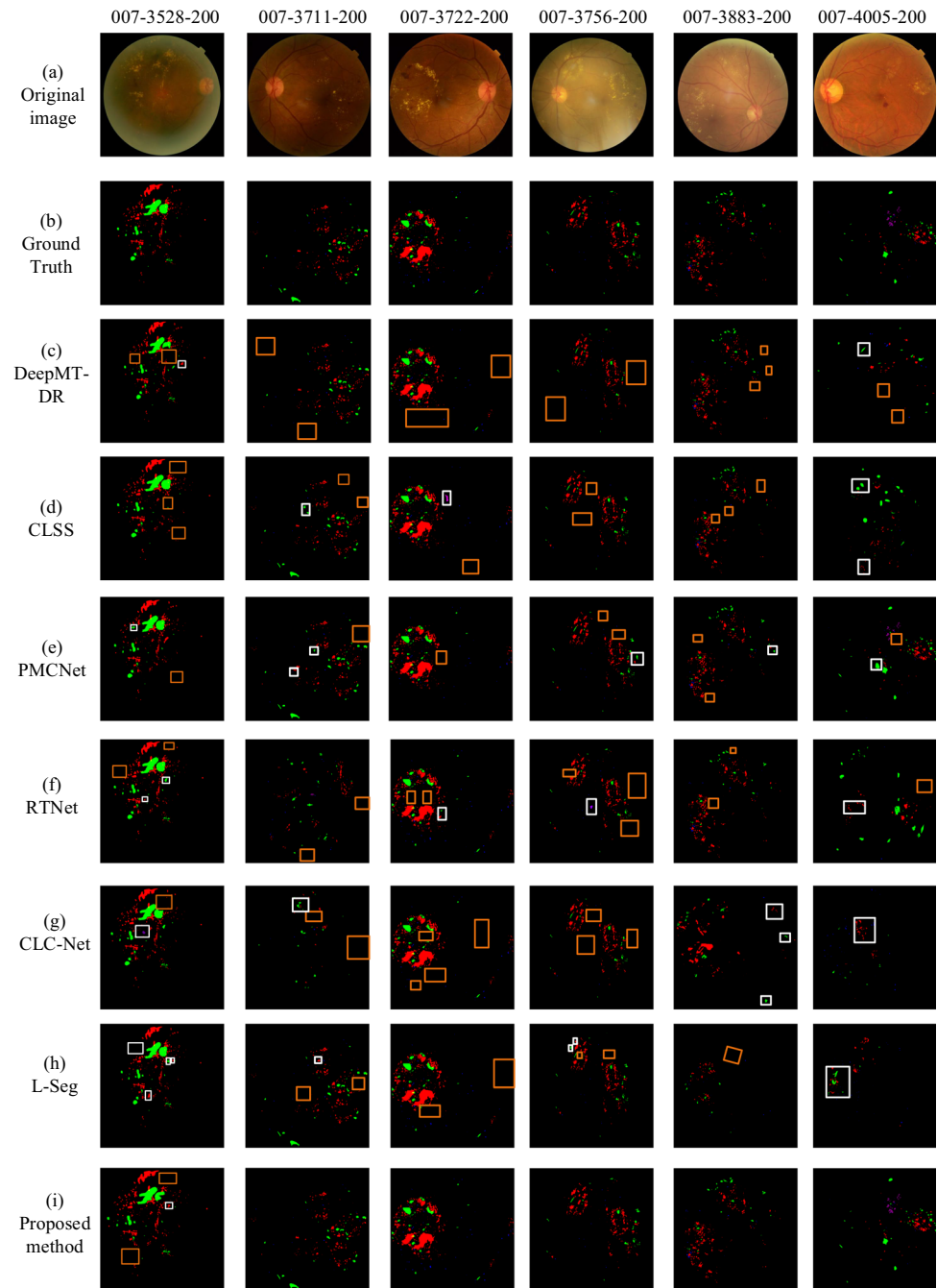
Fig. 9 ROC curve of four lesions segmentation on IDRiD. (a) ROC curve of EX (b) ROC curve of SE (c) ROC curve of HE (d) ROC curve of MA

Fig. 9 (c) and (d), the ROC curve of our method is closer to the upper left boundary, while the ROC curve of MTUNet [10] is closer to the lower right boundary.

4.4.2 Performance on DDR

Figure 10 displays the qualitative comparison of the multiple lesion segmentation results on DDR for eight different methods to verify the effectiveness of the proposed method. The numbers at the top of retinal images in Fig. 10 represent the specific sequence numbers of fundus images in DDR dataset. As can be seen from Fig. 10, the contrast of fundus images in DDR is inconsistent and there are dark spots and vessel artifacts that affect the accuracy of lesion segmentation. CLSS [11] tends to misclassify bright spots as soft exudates and ignored small areas of HE. PMCNet [14] tends to miss small and sparse EX. Although CLC-Net [16] and RTNet [15] can accurately detect small EX and HE, they tend to miss MA. L-Seg [18] tends to misclassify vessel artifacts around the optic disk as HE. The multi-channel loss function in L-Seg cannot completely overcome the adverse interference

Fig. 10 Qualitative comparison results of various networks on DDR. The orange and white bounding boxes represent false negatives and false positives, respectively



of background noise on lesion segmentation. Compared with other existing methods, the edges of the lesion segmentation results are clearer, and the shape and scale are closer to expert manual labels.

The quantitative comparison results of different models on DDR dataset is shown in Table 3. It is observed from Table 3 the segmentation performance of our method is better than that of most current methods, AUC_PR and AUC_ROC of EX, SE, HE and MA reaches 0.8417, 0.7819, 0.6907, 0.4697 and 0.9505, 0.9544, 0.9059, 0.8952, respectively. Compared with PMCNet [14], AUC_PR and AUC_ROC of our method are increased by 29.87%, 38.32%, 2.57%, 27.03% and 6.31%, 13.1%, 5.13%, 5.95%. Compared to M2MRF [40], the *Dice* and AUC_PR of our method are increased by 35.97%, 43.04%, 47.35%, 64.13% and 20%, 25%, 13.6%, 18.51%.

Table 3 Quantitative comparison results of lesion segmentation using different models on DDR. (The best results are shown in bold)

Methods	EX			SE			HE			MA		
	Acc	Dice	AUC	PR	AUC	ROC	Acc	Dice	AUC	PR	AUC	ROC
MTUnet [10]	0.8166	0.8397	-	0.9413	0.7852	0.7983	0.7416	0.8027	-	0.9497	0.7167	0.6923
CLSS [11]	0.8174	0.7412	0.7981	-	0.7983	0.6859	0.7613	0.7519	-	-	0.6958	0.6738
DeepMT-DR [12]	0.8871	0.8459	0.8625	0.8383	0.8732	0.8537	0.8598	0.7984	0.4396	0.7955	0.8147	0.7967
CARNet [13]	0.9738	0.9687	0.8373	0.9429	0.9573	0.9465	0.9457	0.9468	0.4032	0.9331	0.9543	0.9307
PMCNet [14]	0.9235	0.9508	0.5430	0.8874	0.8093	0.8324	0.8098	0.8013	0.3164	0.8546	0.8123	0.8034
RTNet [15]	0.9142	0.9247	0.5671	0.9184	0.8425	0.8352	0.8294	0.9341	0.3656	0.7922	0.8245	0.8167
CLC-Net [16]	0.7139	0.8479	0.7130	0.8117	0.8942	0.7985	0.9012	0.8697	0.5860	0.8493	0.8142	0.8916
L-Seg [18]	0.8965	0.9183	0.5546	0.9432	0.8954	0.8897	0.9053	0.8697	0.3586	0.9369	0.8452	0.8874
LKD [37]	-	0.5900	0.6450	-	-	-	-	0.6150	0.6440	-	-	0.5550
HACDR-Net [38]	-	0.6124	0.6515	-	-	0.5564	-	0.5396	0.5502	-	-	0.2334
MSGDA-Net [39]	-	0.7285	0.6918	-	-	0.3177	-	0.5378	0.3980	-	-	0.3845
M2MRF [40]	-	0.6115	0.6417	-	-	0.5210	-	0.4681	0.5547	-	-	0.2804
DMSF [24]	-	-	0.6014	-	-	-	-	-	0.3652	-	-	0.1194
Proposed	0.9636	0.9712	0.8417	0.9505	0.9621	0.9514	0.9434	0.9416	0.6907	0.9059	0.9325	0.9217
											0.4697	0.8952

The ROC curves of different models on DDR dataset for four kinds of lesions segmentation is shown in Fig. 11. The AUC values of our method on the four lesion segmentation are greater than those of the other four models, and the AUC values of EX, SE, HE and MA segmentation reach 0.9505, 0.9544, 0.9059 and 0.8952, respectively. As shown in Fig. 11(a), the AUC of our method is largest, followed by RTNet [15], and the AUC of CLC-Net [16] is smallest. As shown in Fig. 11(c) and (d), the ROC curve of our method is closer to the upper left boundary, while the ROC curve of RTNet [15] is closer to the lower right boundary.

4.4.3 Performance on FGADR

FGADR is a newly released DR image dataset by Zhou et al. [17] in 2020, so few researchers performed experiments to verify the effectiveness of model on this dataset. Therefore, we replicated the code of five existing network models and performed comparative experiments on FGADR. Fig. 12 shows the visual segmentation results of multiclass lesions on FGADR using different models. The numbers at the top of retinal images in Fig. 12 represent the specific sequence numbers of fundus images in FGADR dataset. As can be seen from Fig. 12,

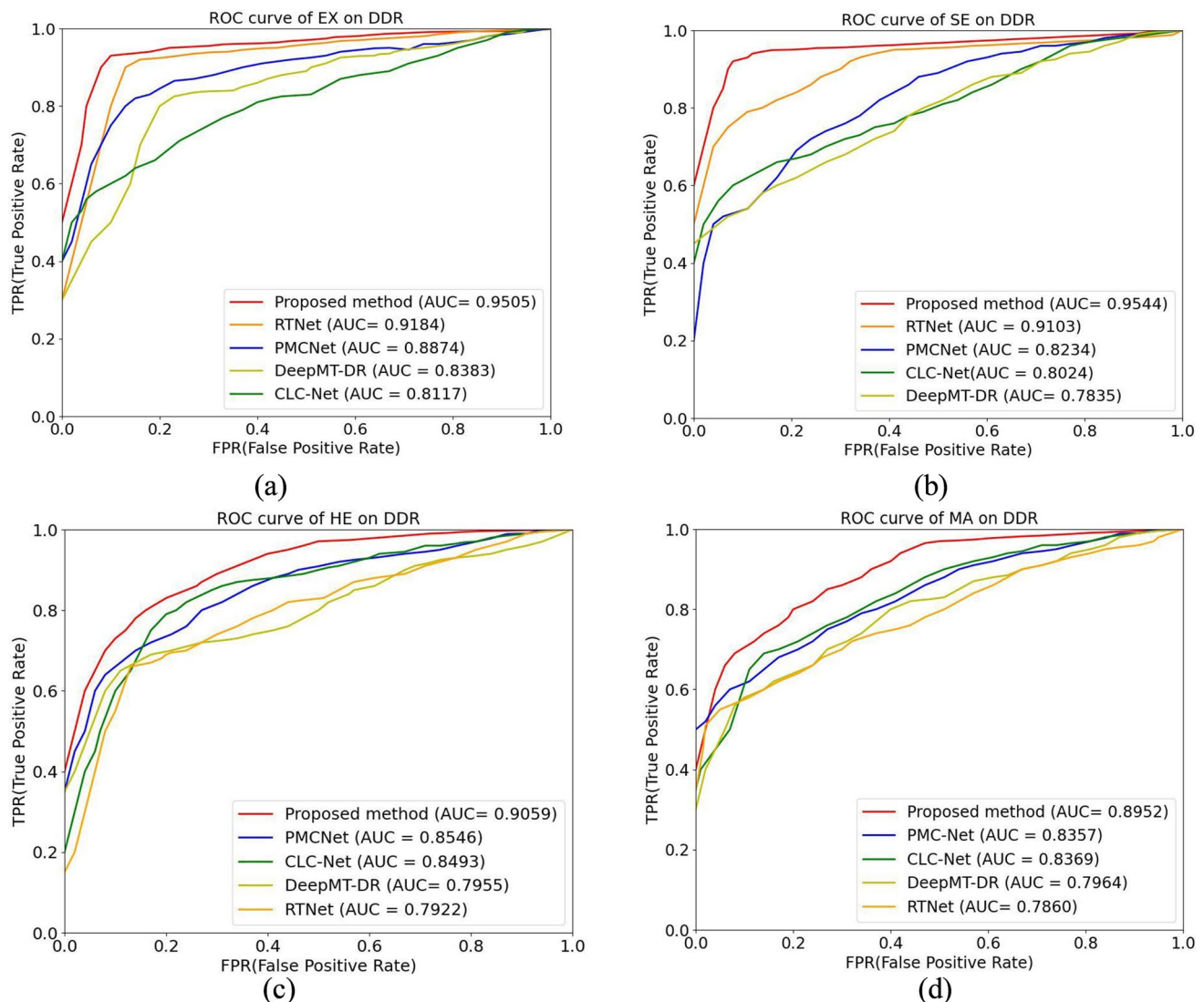
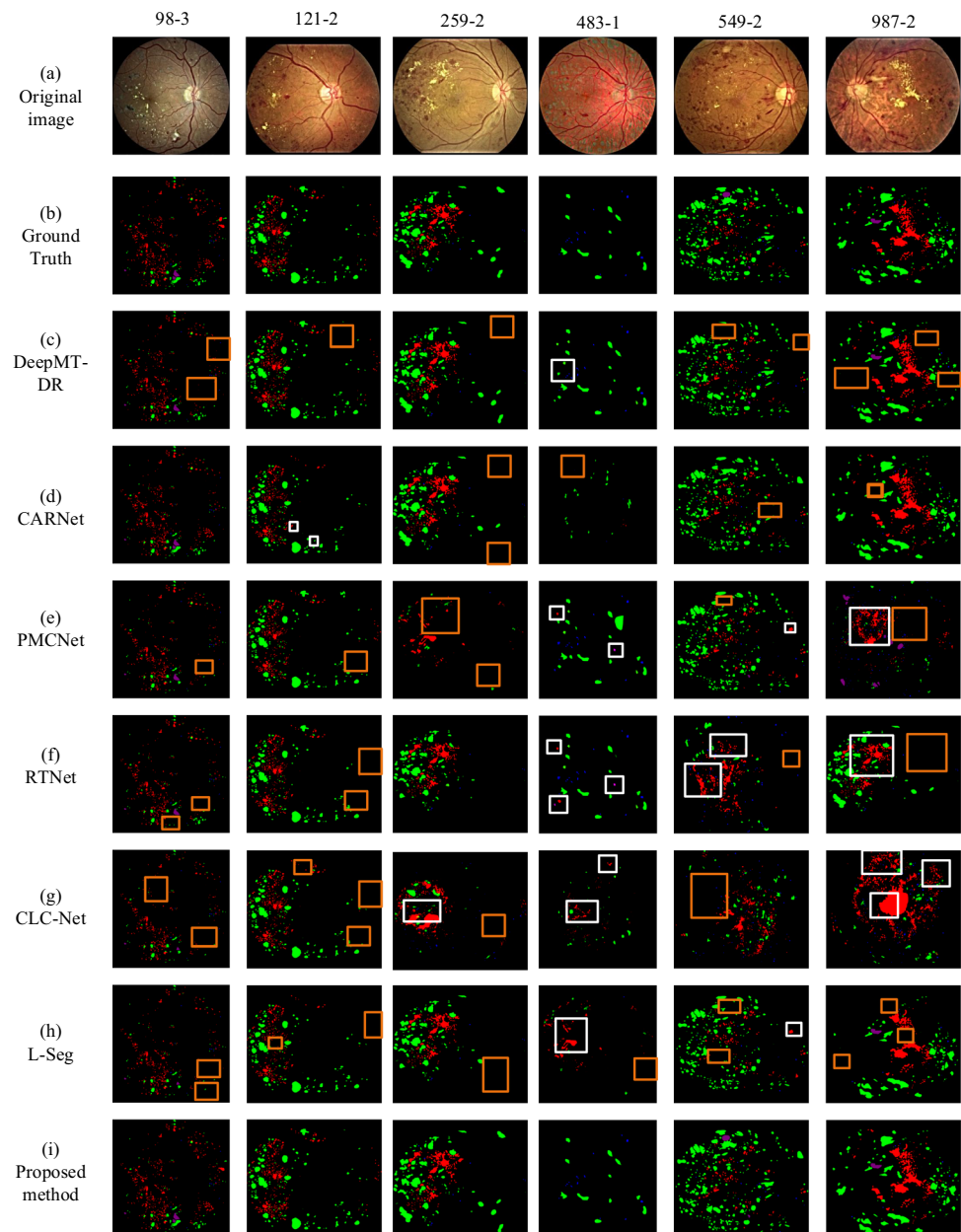


Fig. 11 ROC curves of four lesion segmentation on DDR. (a) ROC curve of EX (b) ROC curve of SE (c) ROC curve of HE (d) ROC curve of MA

Fig. 12 Qualitative segmentation results of different models on FGADR. The orange and white bounding boxes represent false negatives and false positives, respectively



there are multiple dark spot noises and the contrast of fundus images is relatively low. Moreover, the size of multiple lesions is inconsistent and relatively scattered, which adds difficulty to accurate segmentation. DeepMT-DR [12] can segment densely distributed EX and HE, but tends to ignore small and sparse HE. CARNet [13] tends to misclassify small dark spots as HE, resulting in false positive results. PMCNet [14] easily misclassified bright spots as EX and ignored small EX. Although RTNet [15] can detect large areas of HE using global Transformer and local Transformer modules, it easily overlook scattered distribution of HE and EX. CLC-Net [16] is easily affected by background noise and misclassified dark spots as SE. L-Seg [18] tends to misclassify some bright lesions such as artifacts and cotton wool spots as EX. The main reason is that L-Seg can only alleviate the problem of wrong classification to a certain extent by using multi-channel loss, and cannot fundamentally eliminate the adverse effects of other noises.

Compared with other existing approaches, the proposed network achieves clearer and more accurate segmentation results. It can effectively overcome the adverse interference of noises (e.g., spots and artifacts) and accurately segment different types of multi-scale lesions with sparse and dense distributions.

Table 4 presents the quantitative segmentation comparison of different models on FGADR for four types of lesions. As shown in Table 4, the segmentation performance of the proposed network is better than most methods, with *Acc* and *Dice* reaching 0.9475, 0.9020, 0.9102, 0.8012 and 0.9521, 0.9182, 0.9125, 0.8792 for EX, SE, HE and MA segmentations, respectively. Compared with PMCNet [14], *Acc* and *Dice* increased by 2.61%, 9.03%, 1.66%, 6.2% and 1.52%, 2.62%, 3.03%, 4.86%, respectively. Compared to HACDR-Net [38], the *Dice* and AUC_PR of the proposed model are increased by 34.96%, 30.35%, 37.91%, 33.42% and 7.88%, 25.28%, 2.59%, 22.48%.

Figure 13 shows the ROC curves for multiclass lesion segmentation on FGADR of the five models. It can be seen from Fig. 13 that the ROC curve of our method is higher than that of the other four methods in the four types of lesion segmentation, with the highest AUC value, which indicates the proposed method has good superiority in segmentation performance.

4.4.4 Performance on E-ophtha

Figure 14 shows the EX and MA segmentation results of six models on E-ophtha dataset. Compared with the other three datasets, the fundus images have lower contrast, more scattered lesion distribution and smaller sizes. Moreover, MA are so small that occupy only a few pixels, which makes it difficult to distinguish even by the human eyes. The numbers at the top of retinal images in Fig. 14 represent the specific sequence numbers of fundus images in E-ophtha dataset. As shown in Fig. 14, CLSS [11], DeepMT-DR [12] and PMCNet [14] are easily influenced by low image contrast and ignored small EX and MA, which leads to generating false negative results. RTNet [15] and L-Seg [18] are susceptible to background noise interference, and mistakenly detect small dark spots as MA, thus resulting in false positive results. However, the proposed network adopts ResNeXt-50 encoder to extract multiscale lesion features and divides the lesions into three different scales of small, medium and large. Adaptive fusion of lesion visual and textual features is used for accurate segmentation of two types of lesions. Compared with other existing methods, the proposed model can effectively overcome the interference of low contrast and background noise, segmenting sparsely distributed and small lesion areas accurately.

Table 5 shows the quantitative segmentation comparison results of multiclass lesions using different models on E-ophtha. It is observed from Table 5 that the quantitative segmentation performance of our network is better than that of most other existing methods, and *Acc*, *Dice*, AUC_PR and AUC_ROC in EX and MA segmentation reach 0.9367, 0.9278, 0.7997, 0.9562 and 0.9218, 0.9131, 0.6295, 0.9078, respectively. Compared with CLC-Net [16], *Acc*, *Dice*, AUC_PR and AUC_ROC of our method improved by 11.33%, 2.51%, 7.77%, 10.55% and 3.24%, 4.52%, 9.35%, 7.61%, respectively. The *Dice* and AUC_PR of the proposed model are increased by 19.93%, 52.86% and 10.79%, 6.48% when compared to MSGDA-Net [39].

Figure 15 shows the ROC curves for segmentation of EX and MA using five different models on E-ophtha dataset. As shown in Fig. 15, our method outperforms the other four models in both EX and MA segmentation. The ROC curve of our method is highest, followed by RTNet [15], and ROC curve of MTUNet [10] is lowest.

4.5 Ablation study

To verify the effectiveness of each module in TVFNet and its contribution to the segmentation results, extensive ablation studies are carried out to analyze the performance of the backbone network, FEM, MSFM and TLAB. The following ablation studies are conducted on DDR dataset due to the sufficient fundus images.

Table 4 Quantitative comparison results of lesion segmentation using different models on FGADR. (The best results are shown in bold)

Methods	EX			SE			HE			MA		
	Acc	Dice	AUC	PR	AUC	ROC	Acc	Dice	AUC	PR	AUC	ROC
MTUnet [10]	0.8743	0.8689	0.8206	0.8698	0.9354	0.9315	0.9015	0.9321	0.8515	0.9135	0.9067	0.9056
CLSS [11]	0.8752	0.8768	0.8632	0.8956	0.8592	0.9143	0.8862	0.7098	0.7429	0.9012	0.9032	0.9262
DeepMT-DR [12]	0.8782	0.9087	0.7205	0.6613	0.9174	0.9013	0.7495	0.8298	0.8719	0.7988	0.7904	0.7528
CARNet [13]	0.9365	0.9523	0.7985	0.8632	0.9612	0.9351	0.8895	0.9316	0.9079	0.8988	0.9012	0.9048
PMCNet [14]	0.9214	0.9369	0.8023	0.8610	0.8754	0.9152	0.8272	0.9036	0.8879	0.8297	0.8016	0.8199
RTNet [15]	0.9269	0.9352	0.8039	0.8518	0.8997	0.8963	0.8896	0.9063	0.8471	0.7852	0.8671	0.8056
CLC-Net [16]	0.8753	0.9282	0.8269	0.7624	0.7982	0.9063	0.8696	0.8069	0.8761	0.8895	0.8557	0.7674
L-Seg [18]	0.9452	0.9478	0.9205	0.8915	0.9238	0.9389	0.8688	0.8193	0.8937	0.9015	0.9142	0.9213
LKD [37]	-	0.5271	0.8659	0.9236	-	0.6120	0.8874	-	0.5145	0.8256	0.8173	0.9026
HACDR-Net [38]	-	0.6025	0.8869	0.8729	-	0.6147	0.7095	-	0.5334	0.8659	0.8798	0.7059
MSGDA-Net [39]	-	0.7285	-	0.9056	-	0.5378	0.9017	-	0.3845	-	0.8933	0.9012
M2MRF [40]	-	0.6058	-	0.9025	-	0.4381	0.7595	-	0.2804	-	0.8292	0.8912
Proposed	0.9475	0.9521	0.9657	0.9315	0.9020	0.9182	0.9217	0.9102	0.9125	0.8918	0.9069	0.8980

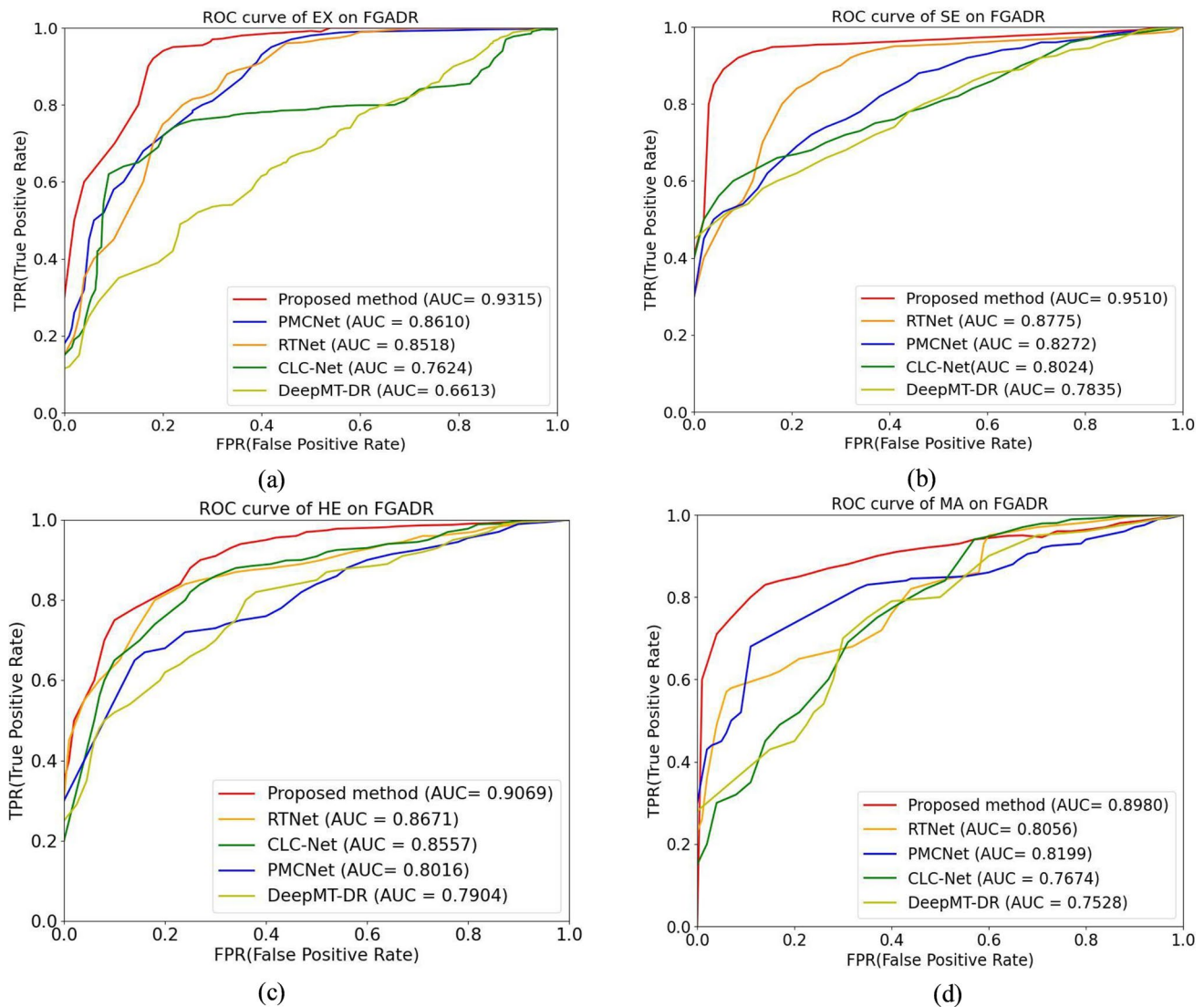


Fig. 13 ROC curves for segmentation of the four types lesions on FGADR. (a) ROC curve of EX (b) ROC curve of SE (c) ROC curve of HE (d) ROC curve of MA

4.5.1 Analysis of backbone

To verify the superiority of ResNeXt-50 backbone, we compare five different backbones in terms of model parameters, testing time and segmentation performance. The performance comparison results of various encoder backbones on DDR are shown in Table 6. As shown in Table 6, although the Xception [42], VGG16 [20] and Inception-V3 [43] have slightly fewer parameters than ResNeXt-50, the *mAUC* value and testing time of our method are better than the other three models. Specifically, the *mAUC* values of our method for four kinds of lesion segmentation are 1.9%, 1.31%, and 3.83% higher than those of Xception, VGG16 and Inception-V3, respectively. Moreover, compared with ResNet-50 [44] and DenseNet-121 [45], our model has higher *mAUC* values and fewer parameters. Considering the above three factors, we select ResNeXt-50 as the backbone for feature extraction.

Fig. 14 Visualization of multiple lesion segmentation results on E-optha for different methods. The orange and white bounding boxes represent false negatives and false positives, respectively. **(a)** Segmentation results of EX **(b)** segmentation results of MA

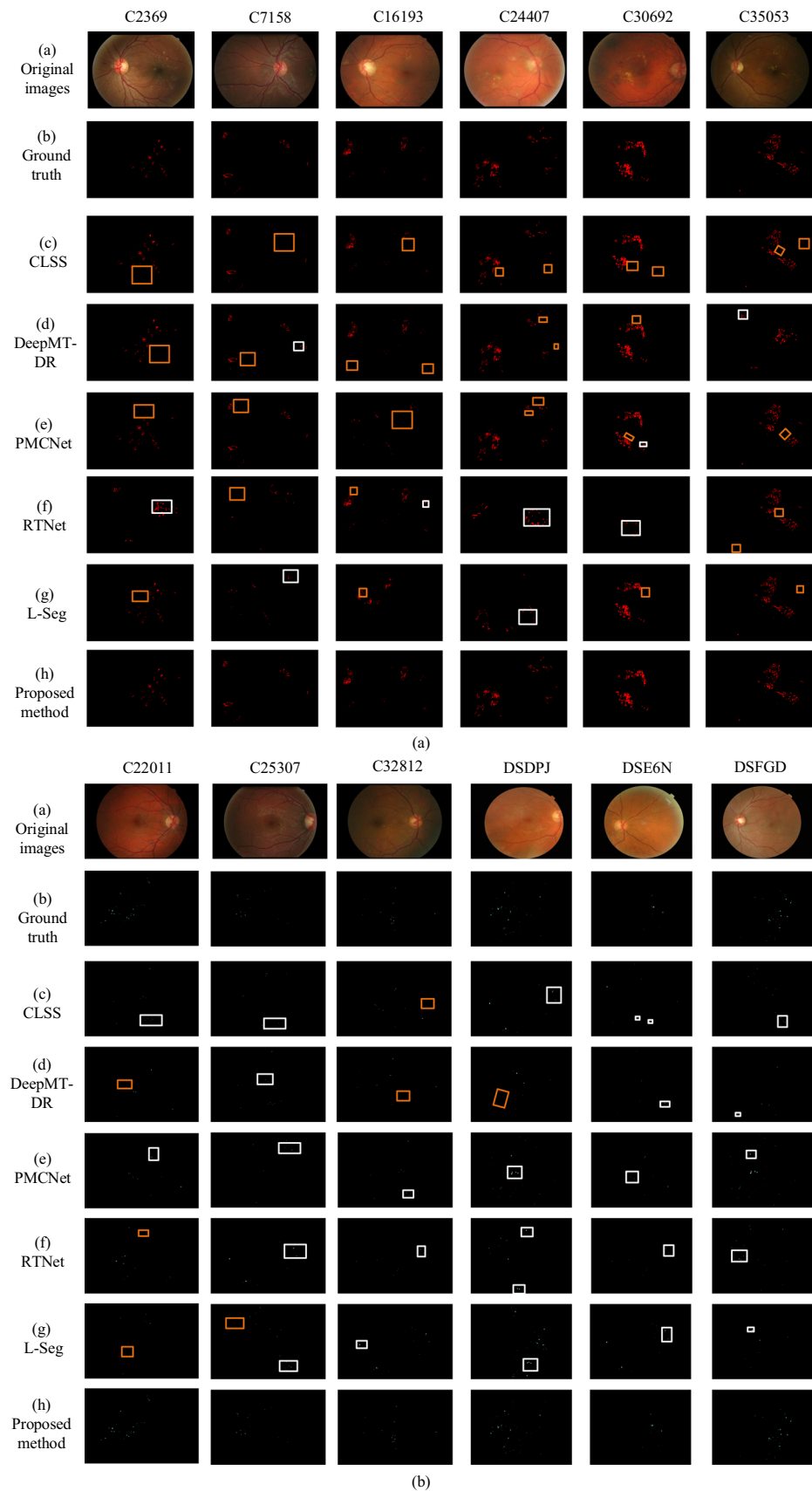
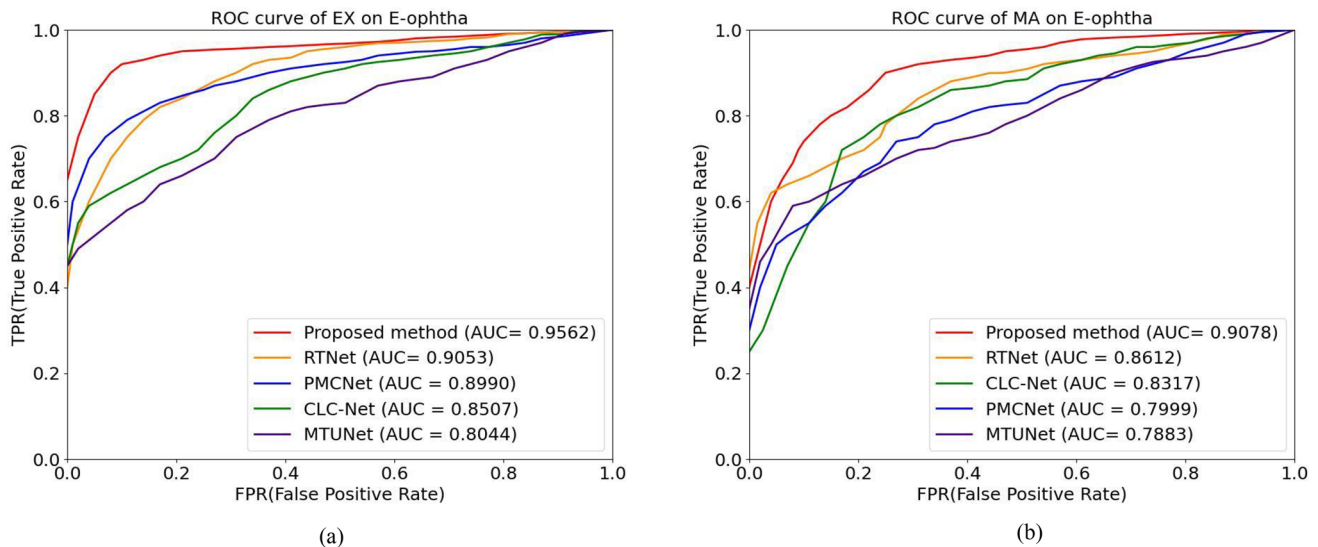


Table 5 Quantitative comparison results of lesion segmentation using different models on E-optha. (The best results are shown in bold)

Methods	EX				MA			
	<i>Acc</i>	<i>Dice</i>	AUC_PR	AUC_ROC	<i>Acc</i>	<i>Dice</i>	AUC_PR	AUC_ROC
MTUnet [10]	0.8971	0.8439	0.4197	0.8044	0.8459	0.8793	0.3987	0.7883
CLSS [11]	0.8864	0.8824	0.5031	-	0.8147	0.9021	0.4062	-
DeepMT-DR [12]	0.8893	0.8952	0.4493	0.8697	0.8594	0.5039	0.5069	0.7036
CARNet [13]	0.9257	0.9198	-	0.9082	0.9058	0.8953	-	0.9017
PMCNet [14]	0.9017	0.9268	0.5120	0.8990	0.9089	0.9187	0.3060	0.7999
RTNet [15]	0.9269	0.9283	0.5146	0.9053	0.9052	0.9127	0.1925	0.8612
CLC-Net [16]	0.8234	0.9027	0.7220	0.8507	0.8894	0.8697	0.5360	0.8317
L-Seg [18]	0.9217	0.9209	0.4171	0.8917	0.9128	0.8713	0.1687	0.8693
DMS [22]	0.8021	0.8319	0.8001	0.8793	-	0.9782	0.2179	0.8419
MLSF-Net [41]	0.9308	0.7963	0.6943	0.9128	0.6971	0.8725	0.5945	0.8972
LKD [37]	-	0.5560	0.6369	0.6050	-	0.5550	0.6020	0.7159
HACDR-Net [38]	-	0.6025	0.6147	-	-	0.2334	0.2450	-
MSGDA-Net [39]	-	0.7285	0.6918	0.5378	-	0.3845	0.5647	0.5098
M2MRF [40]	-	0.6058	0.7417	0.4381	-	0.2804	0.2846	0.4895
Proposed	0.9367	0.9278	0.7997	0.9562	0.9218	0.9131	0.6295	0.9078

**Fig. 15** ROC curve for lesion segmentation on E-optha. **(a)** ROC curve of EX **(b)** ROC curve of MA**Table 6** Comparison results of different backbone networks for the encoder on DDR

Encoder backbone	AUC_ROC				mAUC	Parameters (M)	Test time (s)
	EX	SE	HE	MA			
ResNet-50 [44]	0.8874	0.8429	0.8467	0.8127	0.8474	25.26	0.47
Xception [42]	0.8792	0.8863	0.9028	0.8239	0.8731	22.86	0.81
VGG-16 [20]	0.8819	0.8421	0.8397	0.7396	0.8258	22.47	0.44
DenseNet-121 [45]	0.9004	0.8495	0.8358	0.7986	0.8461	33	0.54
Inception-V3 [43]	0.8972	0.8779	0.8863	0.8537	0.8787	23.7	0.69
ResNeXt-50 [30]	0.9016	0.8975	0.8871	0.8792	0.8914	25	0.39
TVMNet	0.9505	0.9544	0.9059	0.8952	0.9265	30.36	0.14

Table 7 Verify the effectiveness of various modules in the proposed model

Methods	AUC_ROC				mAUC	mAcc	mDice
	EX	SE	HE	MA			
Baseline	0.8891	0.8755	0.8671	0.8522	0.8709	0.8892	0.8993
Baseline+FEM	0.9165	0.9012	0.8931	0.8816	0.8966	0.9213	0.9269
Baseline+MSFM	0.9219	0.9169	0.8967	0.8912	0.9067	0.9261	0.9302
Baseline+TLAM	0.9266	0.9327	0.9006	0.8897	0.9124	0.9278	0.9317
Baseline+FEM+MSFM	0.9305	0.9369	0.9025	0.8904	0.9151	0.9315	0.9322
Baseline+FEM+TLAM	0.9308	0.9572	0.9031	0.8926	0.9159	0.9406	0.9327
Baseline+MSFM+TLAM	0.9310	0.9422	0.9073	0.8966	0.9184	0.9427	0.9315
Baseline+FEM+MSFM+TLAM (Ours)	0.9315	0.9510	0.9069	0.8980	0.9219	0.9440	0.9354

Table 8 Verify the contribution of the triple attention mechanisms in the FEM on the DDR dataset

Methods	AUC_ROC				mAUC	mAcc	mDice
	EX	SE	HE	MA			
Baseline	0.8991	0.8755	0.8671	0.8522	0.8709	0.8892	0.8993
Baseline+CAM	0.9001	0.8789	0.8905	0.8625	0.8789	0.8896	0.9021
Baseline+ SAM	0.9004	0.8806	0.8812	0.8676	0.8861	0.8905	0.9057
Baseline+ PAM	0.9109	0.8904	0.8824	0.8729	0.8893	0.8917	0.9084
Baseline+ SAM+PAM	0.9152	0.8936	0.8869	0.8792	0.8902	0.9052	0.9132
Baseline+ SAM+CAM	0.9128	0.8967	0.8955	0.8736	0.8916	0.9063	0.9187
Baseline+ PAM+CAM	0.9143	0.9005	0.8921	0.8802	0.8924	0.9175	0.9196
Baseline+ PAM+SAM	0.9126	0.9008	0.8926	0.8805	0.8933	0.9198	0.9208
Baseline+ PAM+CAM+SAM	0.9165	0.9012	0.8931	0.8816	0.8966	0.9213	0.9269

4.5.2 Analysis of FEM

To demonstrate the contribution of FEM, MSFM and TLAM, Table 7 displays the quantitative results of various components of our method. *mAcc* and *mDice* represent the average accuracy and Dice value of four types of lesion segmentation, respectively. Baseline represents using only the ResNeXt-50 encoder and decoder for lesion segmentation. As shown in Table 7, *mAUC*, *mAcc* and *mDice* of four types of lesions segmentation improved by 2.57%, 3.21% and 2.76% after adding the FEM on baseline, respectively. It indicates that the feature enhancement module made an important contribution to accurate lesion segmentation. The main reason is that FEM captures spatial attention, channel attention and pixel attention features using dilated convolution and triple attention mechanism to get a more comprehensive and effective lesion feature representation, which helps to enhance dynamic features and maintain feature consistency.

To further analyze the impact of the triple attention mechanism on segmentation results, ablation study was conducted on the contributions of spatial, channel, and pixel attention mechanisms. To ensure fairness in experimental comparison, we used ResNeXt-50 encoder and decoder as baseline. Then we fused different combinations of channel attention, spatial attention, and pixel attention with baseline for lesion segmentation. The experimental results are shown in Table 8.

As shown in Table 8, the AUC of EX, SE, HE, and MA are 0.8991, 0.8755, 0.8671, and 0.8522, respectively, when the FEM is not used. After adding channel attention, spatial attention, and pixel attention to baseline, the *mAUC*, *mAcc* and *mDice* of EX, SE, HE, and MA increased by 0.8%/1.52%/1.84%, 0.04%/0.13%/0.5% and 0.28%/0.64%/0.91%, respectively. Compared with fusing only a single attention mechanism, the lesion segmentation performance of fusing any two attention mechanisms has been improved to a certain extent. The best segmentation results achieved when the baseline is integrated with triple attention mechanism. The AUC_ROC of the four lesion segmentation increased by 1.74%, 2.57%, 2.6% and 2.94% when compared to baseline, respectively. It indicates that spatial, channel, and pixel attention all contribute to the accurate segmentation of diabetic

retinopathy lesions to varying degrees, and the fusion of these three attention mechanisms can achieve the best performance improvement.

In addition, Fig. 16 compares the feature maps of different combinations of three attention mechanism to demonstrate the effectiveness of the FEM more intuitively. The yellow box in Fig. 16 represents the suspicious lesion area. It is difficult for baseline to fully extract lesion features without using the triple attention mechanism. It can capture small-scale lesion information and clearly depict specific lesion areas when integrating channel attention to baseline. It can accurately localize suspicious lesion regions and focus on the spatial location of key lesion regions when fusing spatial attention on baseline, while incorporating pixel attention into baseline can accurately capture lesion points at different scales. The fusion of baseline with the triple attention mechanism can focus on lesion areas, which helps to comprehensively extract different types of multi-scale lesion features in fundus images and provides a strong basis for accurately segmenting lesions at different scales.

4.5.3 Analysis of MSFM

Table 7 shows that the segmentation performance of the four types of lesions improved to varying degrees when added the MSFM to the baseline. The AUC values of EX, SE, HE and MA segmentation increased by 3.28%, 4.14%, 2.96% and 3.9%, respectively. In addition, compared to using only Baseline+FEM, *mAcc* and *mDice* of the four types of lesions were improved by 1.02% and 0.53% when fusing MSFM on the baseline+FEM, respectively. It indicates that MSFM plays an important role in accurately segmenting lesions.

EX and SE are similar in color and structure, and HE and MA are relatively small and have similar brightness and texture in DR images. However, the above ablation experiment results demonstrate that MSFM can integrate low-level lesion detail information and high-level semantic features to significantly enhance discrimination ability of similar lesions for improving the segmentation performance of small-scale lesions.

4.5.4 Analysis of TLAM

Compared with the baseline, the *mAUC*, *mAcc* and *mDice* of the four lesions segmentation increased by 2.1%, 1.58% and 1.24% in Table 7, respectively. Especially, the AUC values of EX and SE increased by 2.5% and 3.52%, respectively. It indicates that TLAM contributes significantly to accurate multiple lesion segmentation. TLAM can effectively enhance the representative lesion features from multiple sources and suppress redundant information by adaptive fusion of text and visual features, which pays more attention to multiscale lesions.

The proposed method achieved the best experimental performance when compared to the baseline, with increases of 5.1%, 5.48% and 3.61% in *mAUC*, *mAcc* and *mDice*, respectively. Moreover, combining any two

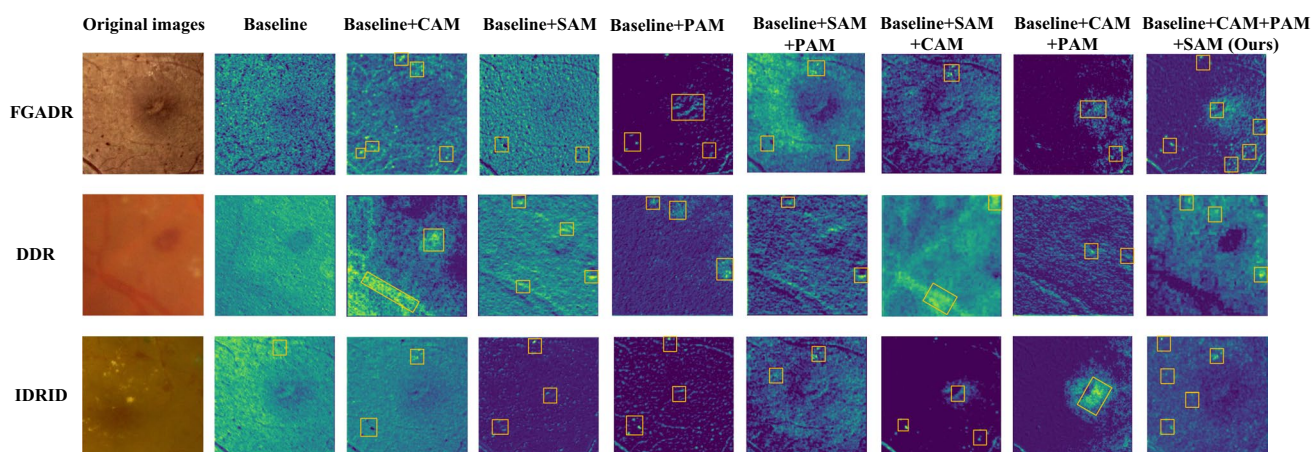


Fig. 16 Lesion feature maps using different attention mechanisms in the feature enhancement module

modules from FEM, MSFM and TLAM resulted in better segmentation performance than only fusing one module with baseline. The aforementioned ablation experiment results demonstrate that TLAM, MSFM and FEM contribute to accurate segmentation of multiple types of lesions to varying degrees. Integrating the three modules with baseline can achieve the best segmentation results.

Moreover, Fig. 17 shows the PR curves of multiclass lesion segmentation from fusing various modules on DDR dataset. As shown in Fig. 17, regardless of the type of lesion segmentation, the PR curve of using only the ResNeXt-50 backbone is the lowest. As TLAM, FEM and MSFM are integrated into ResNeXt-50, the PR curve gradually increases. The PR curve is closest to the upper right corner boundary of the coordinate system when ResNeXt-50 combined with TLAM, FEM and MSFM. The above experimental results can effectively prove the effectiveness of the proposed three modules.

To further explore the effectiveness of byte-pair encoding in the text label attention module, we compared and analyzed the performance differences of multimodal information fusion among three methods: one-hot encoding, word embedding, and byte-pair encoding. As shown in Table 9, compared with using byte-pair coding, the number of computational parameters using one-hot encoding is nearly three times that of byte-pair coding, and

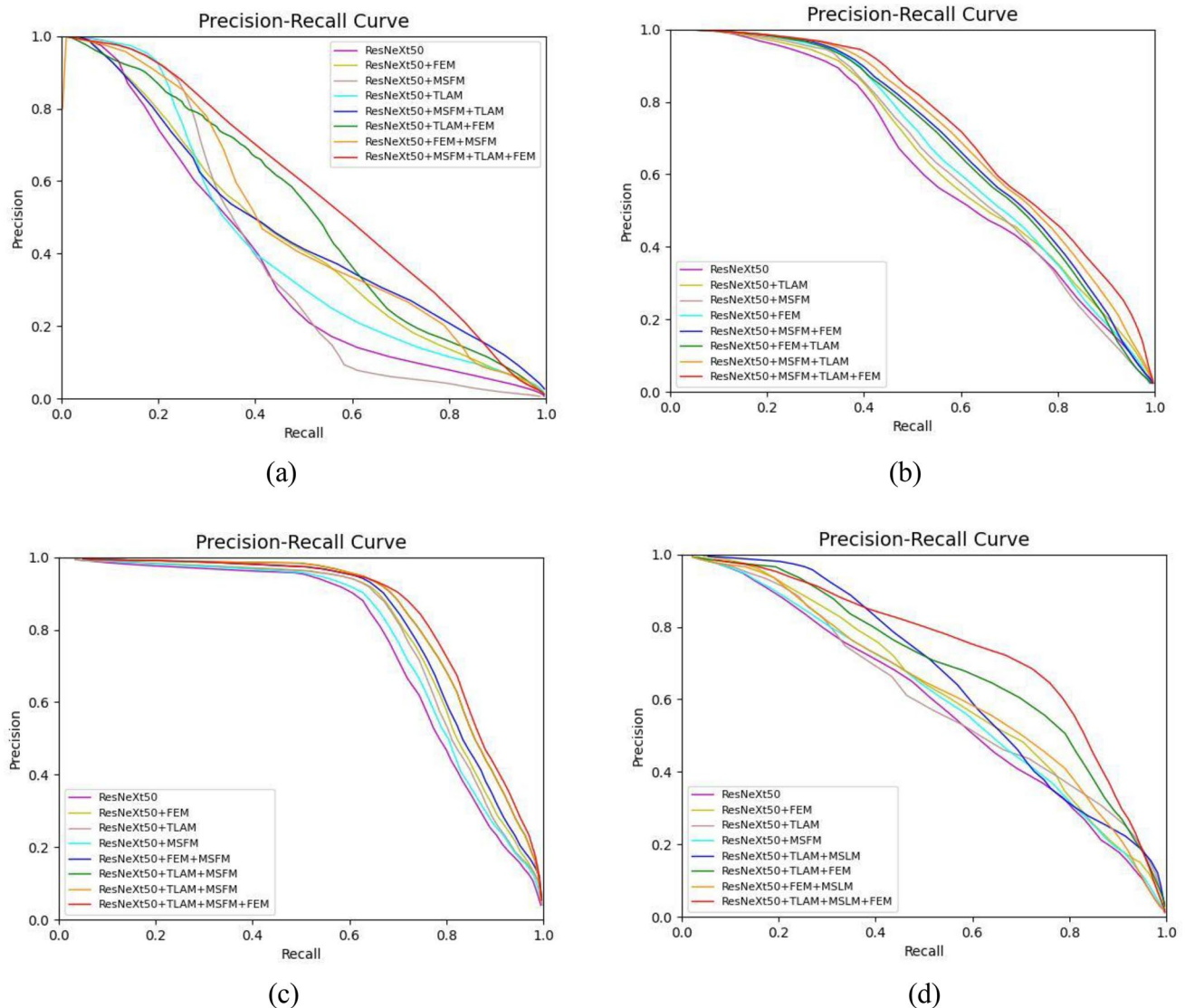


Fig. 17 PR curves for multiclass lesion segmentation integrated with different modules on DDR dataset

Table 9 Verify the effectiveness of the text label attention module

Text encoding methods	Parameters (M)	FLOPs (G)	IDRiD		DDR		E-ophta	
			mDice	mAUC_ROC	mDice	mAUC_ROC	mDice	mAUC_ROC
One-hot	90.9	89.32	0.9029	0.9120	0.9019	0.9125	0.8978	0.9036
Word embedding	29.8	56.98	0.9126	0.9178	0.8968	0.8952	0.8874	0.9158
Byte-pair encoding	30.36	78.36	0.9377	0.9403	0.9464	0.9265	0.9204	0.9320

the mDice and mAUC_ROC on the three datasets are reduced by 3.48%/4.45%/2.26% and 2.83%/1.4%/2.84%, respectively. The main reason is that one-hot encoding converts the scale and type text labels of each lesion into high-dimensional sparse vectors. Each category corresponds to a unique vector position, where the position belonging to the category is 1 and the rest are 0. When the number of categories is large, it leads to high-dimensional sparse vectors, which increases memory and computational complexity.

Although word embedding requires fewer computational parameters for text encoding than byte pair encoding, its segmentation performance is not as good as that of byte pair encoding. The main reason is that word embedding learns low dimensional vectors to represent high-dimensional classification data, which can reduce data dimensionality and improve computational efficiency. In addition, it can preserve similar semantic relationships in labels. The best lesion segmentation results are achieved when byte-pair coding is performed on the number of lesion types and scale labels and element-by-element multiplication with the classification probabilities. The main reason is that byte-pair coding provides a better balance between segmentation granularity and computational efficiency. It adapts to the dataset by creating a vocabulary that effectively represents common character sequences or words, which reduces the complexity of the model and shortens the training time, and ensures better segmentation performance while improving computational efficiency.

4.6 Generalization study

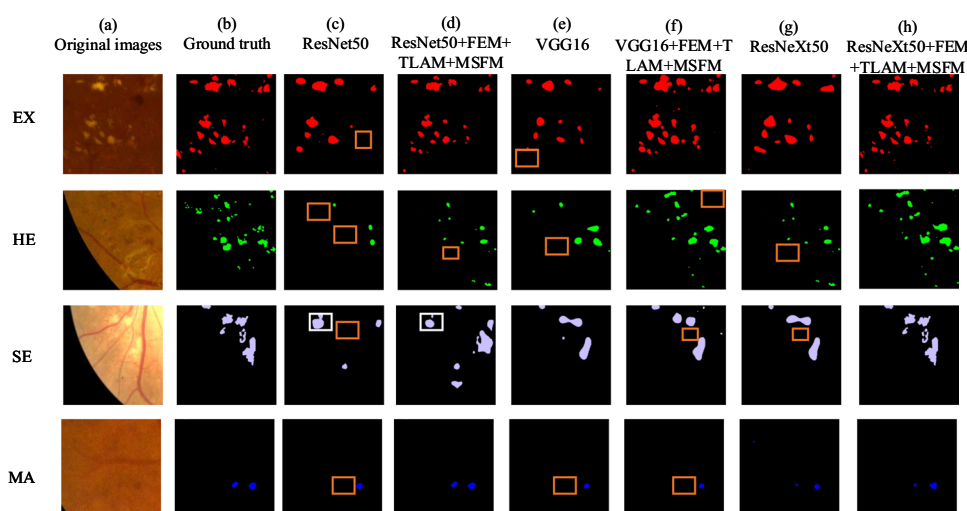
Experiments are conducted on IDRiD, DDR, FGADR, and E-ophta datasets to validate the generalization of the proposed method. Domain transfer between images in different datasets could affect the generalization ability of the model. The main reasons are shown in the following three aspects. (1) Geographic and demographic biases lead to differences in the appearance of fundus images. IDRiD and DDR are from the Indian and Chinese patient populations, respectively, while E-ophta and FGADR are mainly from the European and Asian patient populations. It introduces geographic and race-specific biases due to differences in fundus pigmentation, which influences the generalizability of the models. (2) Labeling standards and disease severity distributions are various among datasets. IDRiD, DDR and FGADR provide fine-grained lesion labeling, whereas E-ophta only labels specific lesions. Inconsistent labeling affects the ability of the model to identify complex cases. In addition, the uneven distribution of disease severity across datasets may lead to insufficient sensitivity of the model to severe lesions. (3) The image acquisition devices are different. The use of multiple fundus cameras (e.g., Zeiss, Canon) in different datasets results in differences in resolution and color contrast of fundus image, which may make the model overly dependent on device-related features.

To verify the generalization performance, we conduct generalization experiments of with CLSS [11] and DeepMT-DR [12] on IDRiD, DDR and E-ophta datasets, i.e., training on one of the datasets and testing on the other two datasets. As shown in Table 10, both the proposed method and the other two comparison methods have certain generalization performance. Specifically, when trained on IDRiD and tested on DDR and E-ophta, the mAcc and mDice of our method reached 0.9012/0.9315 and 0.9132/0.9252, respectively. Compared with CLSS [11], the mAcc and mDice values have increased by 6.11%/0.57% and 2.68%/20.07% on DDR and E-ophta, respectively. Compared with DeepMT-DR [12], the proposed method improved the mAcc and mDice values by 7.58%/6.88% and 1.11%/4.69%, respectively, when trained on E-ophta and tested on DDR and IDRiD.

Although the fundus images in the IDRiD, DDR, and E-ophta datasets come from different countries and ethnicities, with differences in sample size, category distribution, and image appearance, our method can achieve

Table 10 Generalization study of the TVFNet on the IDRiD, DDR and E-optha datasets

Training sets	Testing sets	DDR		E-optha		IDRiD	
	Methods	mAcc	mDice	mAcc	mDice	mAcc	mDice
IDRiD	CLSS [11]	0.7682	0.7132	0.8401	0.9102	0.9258	0.7245
	DeepMT-DR [12]	0.8437	0.8263	0.8505	0.8493	0.9120	0.8116
	Ours	0.8952	0.8695	0.9012	0.8864	0.9315	0.9252
DDR	CLSS [11]	0.8274	0.7413	0.8219	0.8820	0.8319	0.8421
	DeepMT-DR [12]	0.8312	0.8195	0.8826	0.8629	0.8596	0.8892
	Ours	0.9495	0.9125	0.8596	0.9236	0.9014	0.9156
E-optha	CLSS [11]	0.8269	0.8357	0.9028	0.9015	0.8187	0.8692
	DeepMT-DR [12]	0.8096	0.8851	0.8927	0.8907	0.8528	0.8547
	Ours	0.8854	0.8962	0.9128	0.9069	0.9216	0.9016

Fig. 18 Multiclass lesion segmentation results of different backbones integrated with the DDR dataset


better classification performance under the same experimental settings, with mAcc and mDice better than the segmentation performance of the other two methods. The above experimental results show that the proposed method can overcome the adverse interference of different appearance and category distributions of training and testing data, extract key lesion image features and text information for adaptive fusion to promote accurate lesion segmentation, and has strong generalization performance.

Furthermore, extensive experiments were performed on the DDR dataset to analyze the effectiveness of ResNeXt-50 backbone and generalizability of each module more comprehensively. We presented the visual lesion segmentation results on DDR dataset using fusion of the three proposed modules on the ResNet-50, VGG16 and ResNeXt-50 backbones for comparison. Figure 18 shows the EX, HE, SE and MA segmentation results on the patch image from top to bottom. The white and orange boxes represent false positives and false negatives, respectively. As shown in Fig. 18, using only VGG16, ResNet-50 or ResNeXt-50 results in local false positive and false negative results, while integrating the proposed three modules effectively reduces the number of misclassified lesions. It indicates that the three modules contribute to improving the accuracy of lesion segmentation regardless of the type of backbone foundation. Compared to the other backbones, it can achieve better results when integrating ResNeXt-50 with FEM, TLAM and MSFM.

4.7 Computation time

Table 11 shows the calculation time of different models to analyze the time complexity. To ensure a fair comparison, we used the same experimental settings for all models (e.g., the number of training epochs, iteration times

Table 11 The computational performance of different models on the IDRiD, DDR, E-ophta and FGADR datasets

Methods	IDRiD		DDR		E-ophta		FGADR		FLOPs (G)	Parameters (M)
	Training time (h)	Test time (s)	Training time (h)	Test time (s)	Training time (h)	Test time (s)	training time (h)	Test time (s)		
M2MRF [40]	6.2	0.23	6.5	0.21	6.0	0.2	6.3	0.22	77.17	69.93
PMCNet [14]	6.5	0.19	6.6	0.17	6.8	0.19	7.0	0.20	0.04	9.05
RTNet [15]	5.5	0.25	6.7	0.24	5.4	0.22	5.5	0.21	72.41	49.63
MSGDA-Net [39]	7.5	0.278	7.7	0.25	7.4	0.26	7.6	0.24s	27.42	58.75
L-Seg [18]	5.8	0.3	6.0	0.28	5.9	0.31	6.2	0.3	80.78	50.02
Ours	5.6	0.15	6.3	0.14	6.2	0.13	6.1	0.11	78.36	31.36

and learning rate decay) and performed experiments on an Nvidia GeForce RTX 2080Ti GPU. As shown in Table 11, the FLOPs and calculation parameters of our model for segmenting four types of lesions on IDRiD are 78.36G and 31.36M, respectively. The proposed TVFNet takes 5.6 hours to train 200 epochs and only 0.15 seconds to test a retinal image with a resolution of 1024×1024 . Moreover, the FLOPs and computational parameters of our method for segmenting four types of lesions on DDR are 78.37G and 31.36M, respectively. It takes 6.3 hours to train 200 epochs, while testing a fundus image takes 0.14s. The above comparative results show that the computational speed of the proposed method exceeds most other methods and has fewer computational parameters. The training time is less than other existing methods, while the testing time is the shortest. It demonstrates that the proposed method not only achieves good segmentation performance, but also has better computational efficiency.

The main reason is that ResNeXt-50 was used as the encoder backbone to extract multi-scale features of retinal images. ResNeXt uses group convolution to build parallel stacked blocks with the same topology structure, which can effectively shorten training time and improve segmentation accuracy without significantly increasing model parameters. PMCNet [14] uses EfficientNetB0 to extract lesion features and adopted a progressive learning strategy to adjust image resolution and regularization factors, which can accelerate network training to reduce training time. RTNet [15] integrates the global Transformer and local Transformer module with U-Net to achieve multi-type lesion segmentation. However, multi-head self-attention and cross-attention introduce more computational parameters, thereby increasing the training time of model. L-Seg [18] improved VGG16 by removing the last pooling layer and all fully-connected layers, and used weighted fusion of side output features for lesion segmentation. VGG16 adopts shallow network and smaller convolution kernels, which ensures receptive field and reduces the parameters of convolution layers, resulting in shorter training time than that of TVFNet. However, the model comes at the cost of increased testing time and decreased segmentation performance.

5 Discussion

Diabetic retinopathy is a significant cause of blindness worldwide. Therefore, accurate and automatic lesion segmentation is particularly important for early detection and regular screening of diabetic retinopathy. Although there has been some progress in multi-lesion segmentation tasks for diabetic retinopathy in recent years, there are still some difficulties and challenges, such as complex lesion structures, inter-class differences and intra-class similarity. To address the problem of varying lesion scales and types in fundus images, this paper proposes a text and visual attention feature fusion network for multi-lesion segmentation of diabetic retinopathy. First, the original fundus image is input into ResNeXt-50 to extract multi-scale lesion features. Then, FEM is used to focus on important lesion areas using dilation convolution, spatial attention, channel attention and pixel attention mechanisms. Next, TLAM is used to adaptively fuse text label information and visual features as supervision information of decoder. The high-level decoder fuses the output of the low-level decoder, TLAM and FEM for feature upsampling. Finally, MSFM performs a 1×1 convolution and sigmoid operation to generate segmentation results for multiple lesions. Experimental results on four public DR image datasets show that TVFNet has good

accuracy and robustness. It can overcome interference from similar tissues and background noise, which achieves accurate multi-scale lesion segmentation in retinal images of different races.

Many researchers have conducted extensive work on DR segmentation and achieve certain results in recent years. The existing methods are divided into two categories of encoder-decoder and non-encoder-decoder segmentation models. The first method resizes or crops the original images into patches and inputs them into encoder-decoder networks for multi-lesion segmentation. However, these networks may lose detailed features of small-scale lesions during downsampling process. The second method directly inputs whole image into non-encoder-decoder networks and uses different fusion ways for multiscale feature aggregation. Although this method can preserve the detailed features of lesions better, it overlooks the inherent connection between fundus visual features and textual contextual information. Considering the above issues, our model introduces text context information as supervision of the network, combines the text label of the number of lesion types and scale with visual features to fuse multi-source information using element-wise dot product. Moreover, TVFNet learns the spatial attention, channel attention and pixel attention features of lesions for accurate segmentation of different types of multi-scale lesions. It can enhance representative and critical lesion features as well as inhibit irrelevant redundant information for accurate lesion segmentation.

To verify the accuracy and robustness of our method, comparison experiments, ablation studies, and generalization studies were performed on IDRiD, DDR, E-optha, and FGADR datasets. The sufficient quantitative and qualitative comparison results show that the proposed method achieves the mDice and mAUC_ROC of 0.9377/0.9464/0.9057/0.9204 and 0.9403/0.9265/0.9304/0.93320 for the segmentation of four lesions on the IDRiD, DDR, E-optha, and FGADR datasets, respectively. Compared with RTNet [15], the proposed method improved the AUC_ROC of four lesions on IDRiD, DDR, and FGADR by 4.63%/5.73%/5.52%/4.53%, 3.21%/4.41%/11.37%/10.92% and 7.97%/7.35%/3.98%/12.69%, respectively, with significant improvements in the segmentation performance of microaneurysms. The visualization segmentation results of four types of lesions show that our method can overcome the adverse interference of noise such as spots and artifacts in complex background and low contrast, segment different types of multi-scale lesions accurately, and the results are closer to manual annotation. The results of ablation studies show that fusion of FEM, MSFM and TLAM on ResNeXt-50 can help to improve the lesion segmentation accuracy.

The main reason is that this paper classify lesions into three scales based on the number of pixels, and uses the ResNeXt-50 encoder to extract global fundus features. The FEM adopts a triple attention mechanism to extract channel attention, spatial attention, and pixel attention features of the lesion area, to measure the importance of different features and enable the network to focus on the lesion area of interest, which is beneficial for dynamically enhancing feature representation and maintaining consistency. The TLAM encodes the text information of lesion scale and type in byte-pairs and multiplies it element by element with the image mask for adaptive fusion, which is beneficial for enhancing key lesion features and suppressing irrelevant redundant information.

Generalization studies have shown that compared to other existing methods, although there are differences in the appearance and lesion types of training and testing data, our method utilizes FEM and TLAM to fully learn important lesion features with strong robustness, overcoming the adverse effects of domain shift and promoting accurate lesion segmentation. In addition, we compared and analyzed the computation time and parameter count of different models on four datasets. The results show that the computational speed of our method exceeds that of most advanced methods and has fewer computational parameters, while ensuring segmentation accuracy and good computational performance. The main reason is that TVFNet uses ResNeXt as the encoder, which replaces the residual blocks of the previous ResNet three-layer convolution with parallel stacked group convolution residual blocks of the same topology structure to extract multi-level lesion features. It can effectively improve the accuracy of the model without significantly increasing the number of computational parameters.

Although our method achieved good results on most of fundus images in the four publicly available datasets, there are still some limitations. First, TVFNet is difficult to segment MA in fundus images with low contrast. The main reason is that MAs are extremely small and only occupy a few pixels. Moreover, MAs are distributed sparsely and often appear simultaneously with other lesions such as HE, which makes it easy to confuse with

similar lesions. In fact, even clinical ophthalmologists cannot detect small and unclear lesion areas with the naked eye alone. In addition, the proposed model relies on the quality of image masks and text information. Therefore, in the future, we will explore mining prior knowledge from unlabeled data and use self-supervised learning to automatically generate text annotations for lesion segmentation.

Moreover, the proposed model was tested on a limited number of publicly available datasets of DR images, without verifying the generalization of the model on large-scale clinical data. However, it requires to address issues (e.g., data diversity, hardware resource constraints, and regulatory approvals) to generalize the proposed method to real clinical applications. Therefore, in the future, domain adaptation (e.g., generative adversarial networks or style transfer) will be used to reduce the impact due to differences in image acquisition devices. A federated learning framework is constructed to jointly train networks from multiple organizations to cover a wider data distribution without sharing raw data. In addition, a lightweight backbone network (e.g., EfficientNet) is used to compute image features to reduce the burden of hardware resources in hospitals. Moreover, differential privacy or homomorphic encryption is used to process sensitive text data such as patient medical history to ensure privacy security. Furthermore, we plan to strengthen the cooperation with hospitals to verify the robustness of the proposed model on different populations and devices, to accumulate the effective evidence for clinical applications.

6 Conclusion

This paper proposes a text and visual attention feature fusion network for multiple lesion segmentation of diabetic retinopathy. The core idea is to classify the types and scales of lesions using ResNeXt-50, and use TLAM to adaptively fuse the lesion text labels and visual features to focus on representative important lesion features and suppress irrelevant information. The proposed model adopts an encoder-decoder framework, which includes five parts of encoder, FEM, TLAM, MSFM and decoder. First, the encoder uses ResNeXt-50 backbone to extract fundus image features and get the probability of the number of lesion types and size. Then, FEM utilizes triple attention mechanism to enhance the explicit relationships in spatial positions, feature channels and pixels between different types of lesions. Next, TLAM multiplies the classification probability with the text label information in element-wise as supervised information to assist decoder in feature upsampling, which pays more attention to multiscale lesions and suppresses redundant background noises. Finally, MSFM utilizes a series of convolutions, batch normalization and ReLU operations to fuse multi-scale features of lesions through residual connections for lesion segmentation. We conducted comprehensive experiments on four DR image datasets from different countries and ethnicities. The experimental results show that compared with existing methods, the proposed method can overcome other lesion noises and the adverse interference of similar tissues, and accurately segment different types of multi-scale lesions. It demonstrates that adaptive fusion of text information and visual features is helpful for accurate segmentation of multiple lesions in complex backgrounds.

Although the proposed method has achieved good lesion segmentation results on most public datasets, its generalization has not been validated on clinical real data and may be affected by domain transfer. Therefore, we plan to combine domain adaptation and synthetic data augmentation techniques (e.g., StyleGAN generating multi-device images) to improve model generalization, and plan to integrate more regional datasets (e.g., MES-SIDOR-2, APTOS2019) and promote standardized annotation protocols to alleviate data bias issues. In addition, in the future, we plan to extend TVFNet to OCT and OCTA modal fundus images for DR lesion segmentation, use the lesion segmentation information to guide the severity grading of the disease, and develop a deep multi-task learning method that combines lesion segmentation and DR grading.

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